

Nonrelativistic and Relativistic Contributions to Spin-Momentum Locking and Spin Photogalvanic Effect in an Altermagnetic Multiferroic

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Altermagnetic multiferroics provide a platform where non-relativistic spin-momentum locking (SML) coexists with relativistic spin textures, yet disentangling their different origins remains challenging. Here, using first-principles density-functional-theory calculations and symmetry analysis of the multipolar expansion of the spin polarization, we investigate the altermagnetic multiferroic BaCuF₄. Its dominant non-relativistic altermagnetic *d*-wave SML originates primarily from a CuF₆ octahedral rotation coupled to the polar distortion and can therefore be reversed by ferroelectric switching. We show that spin-orbit coupling generates additional spin multipoles and, depending on the Néel-vector orientation, induces spin canting and weak ferromagnetism. By exploiting the group-subgroup relations between spin groups and magnetic space groups, we then distinguish non-relativistic altermagnetic, canting-associated, and purely relativistic contributions to the SML. We further show that this hierarchy is encoded in the nonlinear spin-photogalvanic response, whose spin projection selectively probes contributions of different origins. These results establish a direct connection between the microscopic spin texture and nonlinear spin transport in altermagnetic multiferroics.

I. INTRODUCTION

Altermagnets are a recently identified class of compensated collinear magnets in which the interplay between crystal and magnetic symmetries generates a momentum-dependent spin polarization of the electronic states, known as spin-momentum locking (SML)^{1–4}. In contrast to conventional spin textures, which arise from spin-orbit coupling (SOC), altermagnetic SML is purely nonrelativistic in origin. In the presence of SOC, however, additional symmetry-allowed components of the spin polarization can emerge^{5–8}, giving rise to more complex spin textures that can be naturally described within the framework of spin multipoles^{9,10}.

Altermagnetic multiferroics^{11–20}, in which altermagnetism coexists with ferroelectricity, constitute a particularly interesting class of emerging materials in this context. The ferroelectric polarization originates from a polar distortion that breaks inversion symmetry, creating a noncentrosymmetric environment. In the presence of SOC, this can give rise to Rashba or Weyl-like spin splitting⁶, and even persistent spin textures²¹, coexisting with the nonrelativistic altermagnetic SML of the dominant spin component. SOC may also induce weak deviations from the collinear magnetic state, leading to weak ferromagnetism²². Although these relativistic effects have been shown to be significant in several alter-

magnetic multiferroics^{6,21}, a quantitative understanding of their microscopic origins and relative contributions to the spin texture remains limited.

Probing these distinct contributions requires response functions that are sensitive to both inversion-symmetry breaking and the spin-dependent electronic structure. Nonlinear photocurrents have emerged as powerful probes in ferroelectrics and multiferroics^{23–34}, topological materials^{35–37}, and, more recently, systems with unconventional magnetism, including altermagnets^{38–49}. Among these responses, the spin photogalvanic effect – the generation of a dc spin current by light – directly probes the spin-dependent electronic structure^{42–44,50–59}. However, how the different microscopic contributions to SML are encoded in the spin photogalvanic response of altermagnetic multiferroics has yet to be established.

Here, we address these issues in the prototypical altermagnetic multiferroic BaCuF₄ by combining density functional theory (DFT) calculations and symmetry analysis. We identify three distinct contributions to the SML, classified in terms of spin multipoles: an even-wave, nonrelativistic altermagnetic SML of the dominant spin component; a relativistic even-wave SML of the subdominant spin components; and an odd-wave SML that, in the present case, takes the form of Rashba-like contributions. For specific orientations of the Néel vector, SOC also induces spin canting and a weak-ferromagnetic mo-

ment, which further modify the components of the spin texture. We then establish that the spin photogalvanic tensors reflect the distinct microscopic contributions to SML through their characteristic spin-polarization directions and symmetry properties.

In particular, spin currents polarized along the Néel vector are associated with the nonrelativistic altermagnetic quadrupole, those polarized along the weak-ferromagnetic moment with the canting-induced monopole, and those polarized perpendicular to both with relativistic SOC multipoles. Furthermore, the calculated magnitudes reveal a corresponding hierarchy: the non-relativistic contributions produce the largest photocurrents, followed by those associated with the additional spin polarization emerging in the weak-ferromagnetic state, whereas the purely SOC-induced contributions are generally substantially smaller.

The paper is organized as follows. Section II describes the computational methods. Section III presents the DFT and symmetry analysis of the nonrelativistic and relativistic contributions to SML, together with the resulting spin photocurrents. Finally, Section IV summarizes the main conclusions.

II. COMPUTATIONAL DETAILS

The DFT calculations were performed using the projector-augmented wave (PAW) method, as implemented in the VASP code^{60–62}. The exchange-correlation functional was described within the generalized gradient approximation (GGA) using the Perdew–Burke–Ernzerhof functional for solids (PBEsol)^{63,64}.

The ground-state $Cmc2_1$ structure and the hypothetical nonpolar $Cmcm$ reference structure^{65,66} were optimized by relaxing both the ionic positions and lattice parameters until the Hellmann–Feynman forces on all atoms were below 0.005 eV/Å. The structural optimizations were performed assuming the A-AFM ordering of the Cu magnetic moments⁶⁷, which is experimentally observed at high temperature⁶⁵. Symmetry-adapted mode (SAM) analysis was performed using the ISODISTORT software^{68,69}.

To account for the localized nature of the Cu d electrons, the DFT+ U correction was applied within the Liechtenstein⁷⁰ formalism, using on-site Coulomb and exchange parameters of $U = 7.0$ eV and $J = 0.9$ eV, respectively. These values were chosen consistently with previous work⁶⁷, where they provide an accurate description of the electronic structure. A plane-wave kinetic-energy cutoff of 500 eV was adopted, and the Brillouin zone was sampled using an $8 \times 4 \times 8$ Monkhorst–Pack k -point mesh for band structure calculations and $22 \times 7 \times 1$ for constructing the two-dimensional (2D) energy surface with $k_x = 0$.

The second-order photocurrent responses were evaluated following the methodology of Refs.^{27,36,50}, based on interpolation of the DFT band structure using maxi-

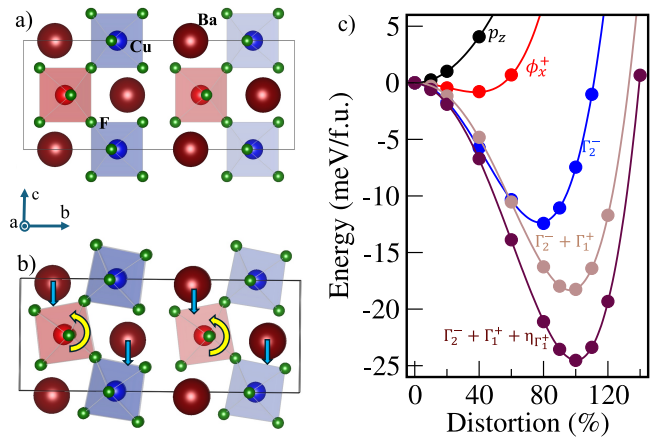


FIG. 1. Schematic representations of (a) the centrosymmetric $Cmcm$ reference structure and (b) the polar $Cmc2_1$ ground-state structure of $BaCuF_4$. Brown, red/blue, and green spheres denote Ba, Cu, and F atoms, respectively. Red and blue Cu atoms indicate oppositely oriented magnetic moments in the A-type antiferromagnetic configuration. Yellow curved arrows indicate the rotations of the CuF_6 octahedra about the x axis, while blue arrows indicate the polar displacements of the Ba atoms along z . (c) Total energy as a function of the amplitudes of the different structural distortions connecting the $Cmcm$ reference structure to the $Cmc2_1$ ground state. The energy of the undistorted $Cmcm$ structure is set to zero.

mally localized Wannier functions (MLWFs)^{71,72}, as implemented in the WANNIER90 package⁷³. Cu d and F p orbitals were included in the Wannierization, resulting in 136 spinor bands in the presence of SOC. The Wannier-interpolated band structures are reported in the Supplemental Material (SM). The photoresponses were evaluated on a dense $100 \times 100 \times 100$ k -point grid, for which the photoconductivities are well converged, with no appreciable changes upon further increasing the grid density.

We use the conventional spin-current definition based on the symmetrized product of the spin and velocity operators, following Ref.⁵⁰. The calculated response is rescaled by $2e/\hbar$ and reported in charge-current-equivalent units.

III. RESULTS

A. Structural properties and electronic structure

$BaCuF_4$ belongs to the $BaMF_4$ family of compounds^{67,74} and crystallizes in the polar orthorhombic space group $Cmc2_1$ (No. 36). The crystal structure consists of layers of corner-sharing CuF_6 octahedra separated by Ba sheets and stacked along the y direction, as shown in Fig. 1(a) and (b). The system is therefore quasi-two-dimensional in the xz plane and exhibits a spontaneous electric polarization along the z

direction ($\mathbf{P} \parallel z$).

To understand the origin of the polar ground state, we performed a symmetry-mode analysis with respect to the hypothetical centrosymmetric $Cmcm$ reference structure⁶⁷. The corresponding energy landscape is shown in Fig. 1(c). The distortion from $Cmcm$ to $Cmc2_1$ can be decomposed into the irreducible representations (IRs) Γ_2^- and Γ_1^+ , together with the associated strain $\eta_{\Gamma_1^+}$. The latter comprises tensile strains of 0.15% and 0.97% along the a and b directions, respectively, and a compressive strain of -0.12% along c .

The Γ_2^- mode comprises two symmetry-adapted distortions: a polar displacement p_z , mainly involving the Ba atoms and responsible for the electric polarization along z ⁶⁷, and an in-phase rotation ϕ_x^+ of the CuF_6 octahedra about the x axis. The two distortions make opposite contributions to the total energy, as shown by the black and red curves, respectively. While the polar mode p_z alone increases the energy of the reference structure, the octahedral rotation ϕ_x^+ lowers it.⁷⁵ Their combination (blue curve) yields a further energy gain and drives the instability of the $Cmcm$ phase, consistent with the unstable Γ -point phonon reported in Ref.⁶⁷.

The Γ_1^+ mode is a fully symmetric distortion involving displacements of Ba, Cu, and F atoms, with the latter two occurring primarily within the CuF_6 layers. It shortens the in-plane Cu–F bonds, resulting in a 2.6% reduction in the volume of the CuF_6 octahedra. Since it does not lower the symmetry of the $Cmcm$ phase, it emerges as a secondary structural response during relaxation of the primary Γ_2^- distortion. Nevertheless, it provides an additional energetic stabilization, as evidenced by the further reduction of the total energy, as shown by the brown curve. Finally, upon inclusion of the strain $\eta_{\Gamma_1^+}$, the system reaches the fully relaxed $Cmc2_1$ ground state, which lies approximately 25 meV/f.u. below the centrosymmetric $Cmcm$ reference phase.

Besides these collective distortions, the CuF_6 octahedra exhibit a local Jahn–Teller (JT) distortion associated with the nominal d^9 configuration of Cu^{2+} . In the approximately octahedral crystal field generated by the surrounding F ligands, the Cu d states split into lower-energy t_{2g} and higher-energy e_g manifolds. The t_{2g} manifold is fully occupied, whereas the e_g manifold hosts a single hole and is therefore JT active. The resulting distortion is already present in the centrosymmetric $Cmcm$ reference structure and persists in the polar $Cmc2_1$ phase; it is therefore distinct from the collective symmetry-breaking distortions driving the $Cmcm \rightarrow Cmc2_1$ transition.

In the relaxed $Cmc2_1$ structure, the JT distortion elongates the Cu–F bonds along the crystallographic a direction to 2.26 Å, compared with an average length of 1.89 Å for the remaining Cu–F bonds. The resulting lifting of the e_g degeneracy places the unoccupied minority-spin state at the bottom of the conduction manifold. It forms the isolated, weakly dispersive band visible approximately 2.7 eV above the valence-band maximum in

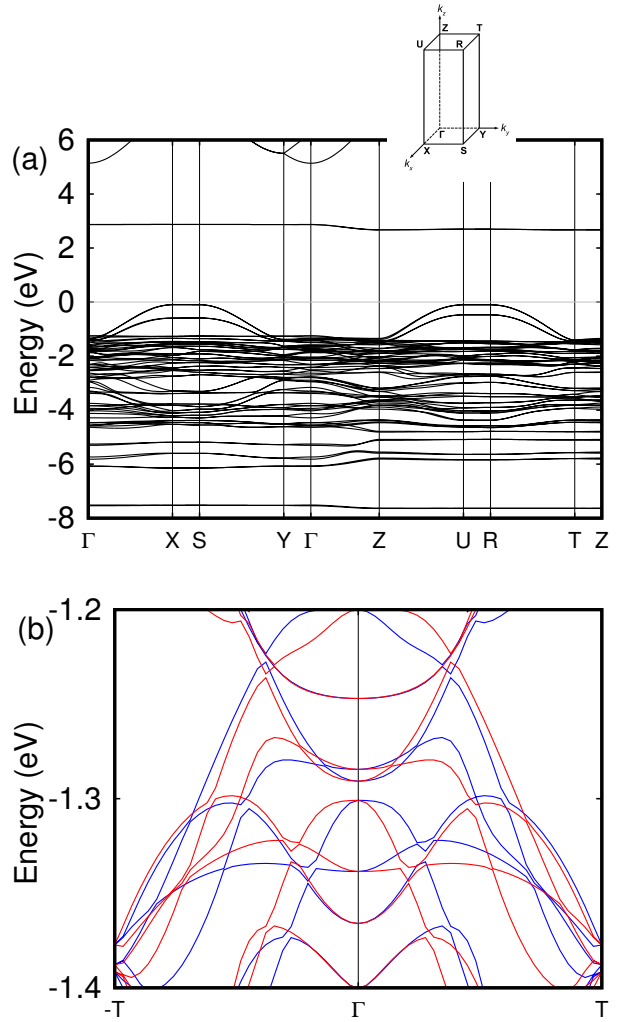


FIG. 2. (a) Nonrelativistic band structure of BaCuF_4 along the high-symmetry path of the orthorhombic BZ shown in the inset. The isolated conduction band, approximately 2.7 eV above the valence-band maximum, originates from the unoccupied Cu e_g states. (b) Nonrelativistic spin splitting along the $-T \leftarrow \Gamma \rightarrow T$ direction of the BZ. Red and blue indicate opposite signs of the spin splitting.

Fig. 2(a), thereby defining the insulating gap. As shown in the SM, the inclusion of SOC leaves the band gap essentially unaffected.

B. Nonrelativistic spin-momentum locking

The centrosymmetric $Cmcm$ reference structure does not exhibit altermagnetism in any of the A-, C-, or G-type magnetic configurations compatible with the unit cell (within the unit cell)⁶⁷. We therefore focus on the polar $Cmc2_1$ phase. We consider throughout this work the A-type antiferromagnetic configuration, which realizes the altermagnetic phase of BaCuF_4 . The A-, G- and C-type orderings are nearly degenerate in energy⁶⁷. In the absence of SOC, the band structure in Fig. 2(b) displays a spin splitting along the $\Gamma = (0, 0, 0) \rightarrow T = (0, \frac{1}{2}, \frac{1}{2})$

Magnetic Configuration	M ₁	M ₂	M ₃	M ₄	M _{tot}	m _a	m _b	m _c
N $\parallel$ <i>x</i>	(m _a ,0,0)	(-m _a ,0,0)	(m _a ,0,0)	(-m _a ,0,0)	(0,0,0)	0.796	0	0
N $\parallel$ <i>y</i>	(0,m _b ,m _c)	(0,-m _b ,m _c)	(0,m _b ,m _c)	(0,-m _b ,m _c)	(0,0,4m _c)	0	0.794	0.020
N $\parallel$ <i>z</i>	(0,-m _b ,m _c)	(0,-m _b ,-m _c)	(0,-m _b ,m _c)	(0,-m _b ,-m _c)	(0,-4m _b ,0)	0	0.046	0.785

TABLE I. Magnetic moments along different directions of the Néel vector (**N**). m_a, m_b and m_c are reported in μ_B . When the Néel vector points along the *y*-axis, weak ferromagnetism appears along *z*, and when it is along the *z*-axis, weak ferromagnetism occurs along *y*.

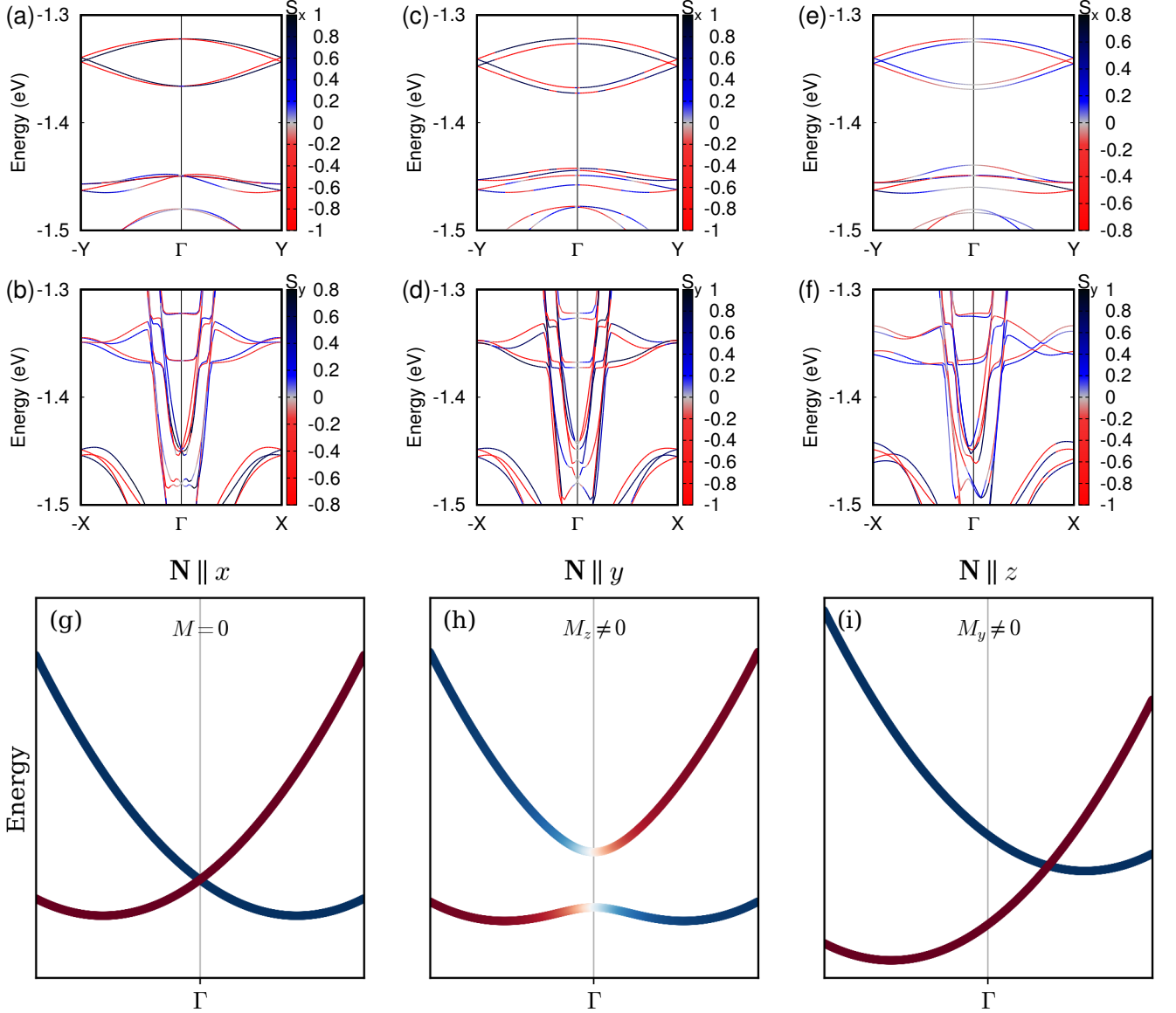


FIG. 3. Rashba-like contributions to the SML for the three Néel-vector orientations: (a),(b) $\mathbf{N} \parallel x$; (c),(d) $\mathbf{N} \parallel y$; and (e),(f) $\mathbf{N} \parallel z$. For $\mathbf{N} \parallel x$, the collinear magnetic state preserves the twofold degeneracy at Γ , whereas for $\mathbf{N} \parallel y, z$, the SOC-induced weak-ferromagnetic moment lifts it. For $\mathbf{N} \parallel y$, $\mathbf{M} \parallel \mathbf{P} \parallel z$, and a gap opens at the crossing. For $\mathbf{N} \parallel z$, $\mathbf{M} \parallel y$ is perpendicular to $\mathbf{P}$, shifting the crossing away from Γ and producing a nonreciprocal dispersion. Panels (g)–(i) schematically illustrate the corresponding evolution of the Rashba-split bands. Red and blue denote opposite signs of the spin polarization.

direction of the Brillouin zone (BZ), shown in the inset of Fig. 2. The splitting is odd in both k_y and k_z , consistently with previous studies¹². More generally, it

vanishes on the $k_y = 0$ and $k_z = 0$ planes, whose intersection defines the nodal line along k_x . The resulting nonrelativistic altermagnetic spin texture therefore has

d -wave character.

The emergence of altermagnetism is closely tied to the structural distortions stabilizing the $Cmc2_1$ phase. The nonrelativistic spin splitting originates primarily from the octahedral rotation mode ϕ_x^+ , whereas the ferroelectric polarization is associated with the polar displacement p_z . These two phenomena are nevertheless coupled: reversing the polarization reverses the sense of the CuF_6 octahedral rotations and, consequently, the sign of the altermagnetic spin splitting in the oppositely polarized $Cmc2_1$ state, in agreement with previous reports^{12,76}.

This coupling between ferroelectricity and altermagnetism is known as the altermagnetoelectric effect¹². Similar mechanisms have also been proposed in perovskite superlattices^{76,77} and rutile compounds⁷⁸. By enabling electrical control of spin polarization, the altermagnetoelectric effect offers a promising platform for reconfigurable spintronic devices.

C. Relativistic contributions to the spin-momentum locking

SOC preserves the nonrelativistic altermagnetic SML of the dominant spin component while generating additional spin-texture contributions. It also determines the magnetic anisotropy: consistent with previous work, the magnetic easy axis is predicted to lie along y . Nevertheless, to understand how relativistic effects reshape the electronic and magnetic structure, we consider all three principal orientations of the Néel vector, $\mathbf{N}$, while retaining the A-type magnetic order throughout.

For $\mathbf{N} \parallel x$, the magnetic order remains collinear, and the system therefore retains a purely altermagnetic character. The dominant spin-polarization component, S_x , is most clearly visible along the Γ -T direction in the BZ, but vanishes along the Γ -X and Γ -Y, where the relativistic contributions can therefore be isolated.

Along these directions, the bands split away from the doubly degenerate Γ point and develop a characteristic camel-back dispersion. The SOC-induced in-plane spin texture has finite S_x along $\bar{Y}$ - Γ - Y and S_y along the orthogonal $\bar{X}$ - Γ - X direction, as shown in Fig. 3(a),(b). Both components reverse sign across Γ , consistent with a spin splitting linear in momentum, and exhibit nodal planes at $k_y = 0$ for S_x and $k_x = 0$ for S_y . These features identify a Rashba-like spin texture arising from the interplay of SOC and the inversion-symmetry breaking associated with the ferroelectric polarization.

Beyond this Rashba-like in-plane spin texture, the DFT calculations additionally reveal an out-of-plane component S_z , with nodal planes at $k_x = 0$ and $k_y = 0$. This therefore constitutes a relativistic d -wave contribution, distinct from the dominant nonrelativistic altermagnetic d -wave texture.

For $\mathbf{N} \parallel y$ and $\mathbf{N} \parallel z$, the nonrelativistic spin polarization is likewise preserved and remains parallel to the Néel vector, with S_y and S_z as the respective dominant

components. In both configurations, however, SOC induces spin canting and a weak-ferromagnetic moment $\mathbf{M}$, directed along z for $\mathbf{N} \parallel y$ and along y for $\mathbf{N} \parallel z$. The system is therefore no longer purely altermagnetic. The calculated atomic magnetic moments and net magnetization per magnetic unit cell are reported in Table I.

The Rashba-like contribution to the spin-polarization persists for both these Néel-vector orientations, producing S_x and S_y components analogous to those found for $\mathbf{N} \parallel x$. However, weak ferromagnetism lifts the double degeneracy of the band crossing at Γ , as expected for a Rashba system subject to an exchange field⁷⁹. The resulting modification of the band structure depends on the orientation of the weak-ferromagnetic moment relative to the electric polarization. The evolution of the Rashba-split bands for the three orientations of the Néel vector is shown in the sketch of Fig. 3(g)-(i).

For $\mathbf{N} \parallel y$, when $\mathbf{M} \parallel \mathbf{P} \parallel z$, a finite S_z develops and a gap opens at the band crossing, as shown in Fig. 3 (c), (d). In contrast, for $\mathbf{N} \parallel z$, $\mathbf{M} \parallel y$ is perpendicular to $\mathbf{P}$, and the band crossing shifts away from Γ . This gives rise to a non-reciprocal band structure, recently identified as a general phenomenon in multiferroics³², whereby the electronic dispersion becomes asymmetric under momentum reversal about certain high-symmetry points of the BZ. For the present $\mathbf{N} \parallel z$ configuration, $\varepsilon_n(\mathbf{k}) \neq \varepsilon_n(-\mathbf{k})$ for $\mathbf{k} \parallel \mathbf{M} \times \mathbf{P}$, as shown in Fig. 3(e), (f). This behaviour is analogous to that found for EuO in the MF-I configuration discussed in Ref.³².

Finally, these Néel-vector configurations exhibit additional S_x contributions, which can be distinguished from the Rashba-like component by their nodal-plane structure. Specifically, for $\mathbf{N} \parallel y$, their nodal planes are located at $k_x = 0$ and $k_z = 0$, whereas for $\mathbf{N} \parallel z$, they occur at $k_x = 0$ and $k_y = 0$. These contributions thus have d -wave symmetry but are relativistic in origin.

In summary, the DFT calculations reveal several SOC-induced contributions to the spin texture beyond the non-relativistic altermagnetic component. They can be classified according to their symmetry, as discussed next.

D. Symmetry analysis of the relativistic spin-momentum locking

The spin polarization near the Γ point can be expanded in powers of the crystal momentum as

$$S_i(\mathbf{k}) = \sum_{n=0}^{\infty} Q_{i,\alpha\beta\gamma\dots}^{(n)} k_\alpha k_\beta k_\gamma \dots \quad (1)$$

where $i, \alpha, \beta, \gamma = x, y, z$ and summation over repeated indices is implied⁸⁰. The coefficients $Q_{i,\alpha\beta\gamma\dots}^{(n)}$ are the components of rank- $(n+1)$ spin multipole tensors, illustrated in Fig. 4. Specifically, the zeroth-order term ($Q^{(0)}$) corresponds to a spin monopole and represents a momentum-independent spin polarization associated with a net magnetization or weak ferromagnetism. The first-order terms

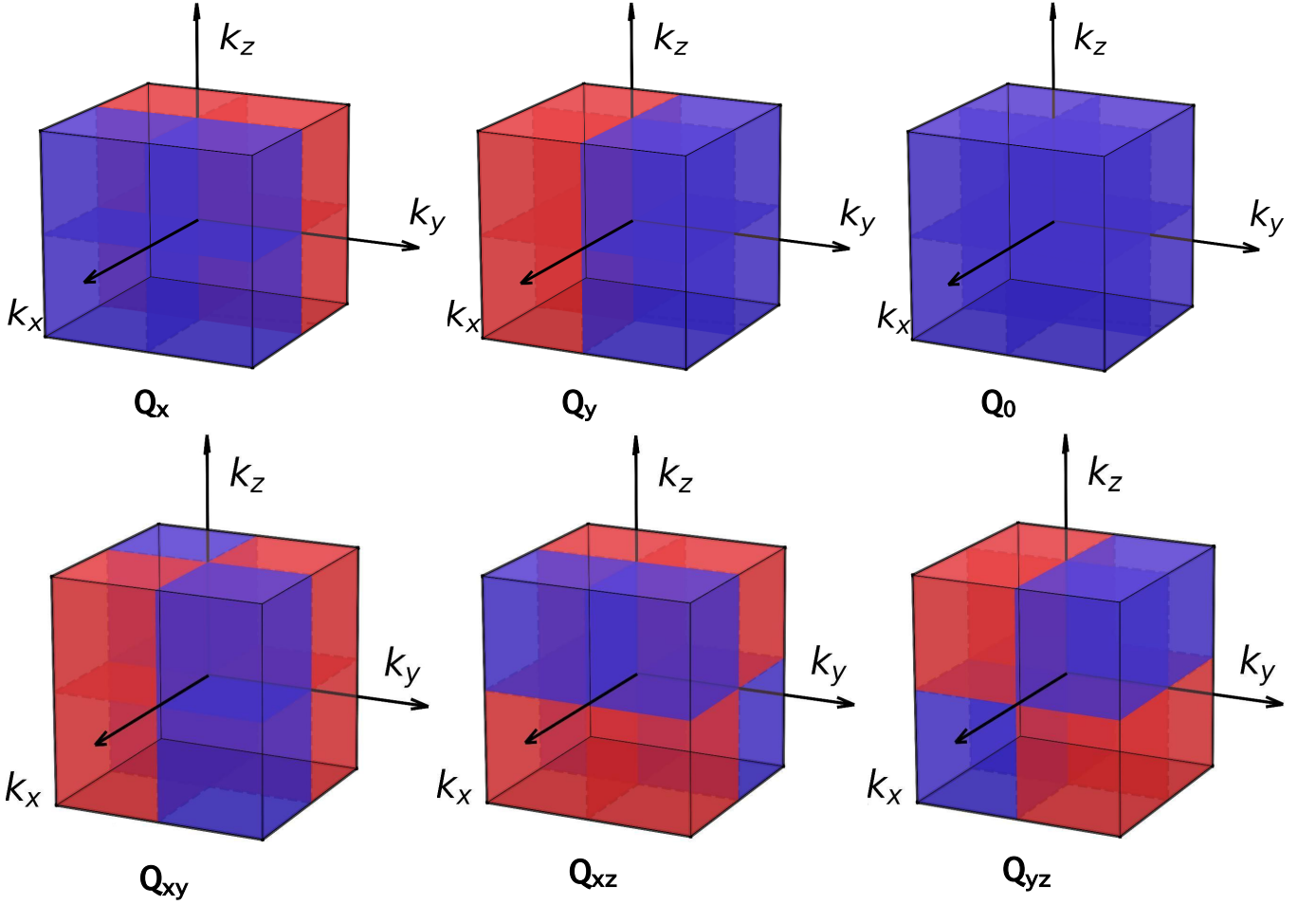


FIG. 4. Momentum-space structures of the monopole Q_0 , dipoles Q_x and Q_y , and quadrupoles Q_{xy} , Q_{xz} , and Q_{yz} relevant to the spin-momentum locking in orthorhombic BaCuF_4 . Red and blue denote regions with opposite signs of the spin polarization.

	Spin components $S_i(\mathbf{k})$		
Néel vector	S_x	S_y	S_z
$\mathbf{N} \parallel x$	Q_y , Q_{yz}	Q_x , Q_{xz}	Q_{xy}
$\mathbf{N} \parallel y$	Q_y , Q_{xz}	Q_x , Q_{yz}	Q_0 , $[Q_{xx}, Q_{yy}, Q_{zz}]$
$\mathbf{N} \parallel z$	Q_y , Q_{xy}	Q_0 , Q_x , $[Q_{xx}, Q_{yy}, Q_{zz}]$	Q_{yz}

TABLE II. Allowed terms of the expansion of band spin-polarization. Purple-coloured terms are those that would survive for collinear spin groups, while orange-coloured ones are those allowed by coplanar spin groups - relevant for the second and third rows. When SOC is included and MSG applies, then all terms appear. Without SOC and for the collinear phases, only the purple-coloured terms survive, while both orange-coloured and purple-coloured terms are allowed for the coplanar (weak-ferromagnetic) cases without SOC. Terms enclosed in square brackets cannot be directly identified from the DFT calculations since they cannot be easily disentangled by Q_0 .

$(Q_x^{(1)}, Q_y^{(1)}, Q_z^{(1)})$ are spin dipoles, characterized by a linear dependence on crystal momentum and a single nodal plane, encompassing Rashba- and Dresselhaus-like spin textures. The second-order terms $(Q_{\alpha\beta}^{(2)})$ are spin quadrupoles, characterized by a quadratic momentum dependence; the off-diagonal components exhibit two or-

thogonal nodal planes.

The allowed spin multipoles depend on the symmetries of both the spin and spatial degrees of freedom. Without SOC, these are described by spin groups; with SOC, spin and real-space transformations are coupled and the relevant symmetries are the magnetic space groups (MSGs). For $\mathbf{N} \parallel x, y, z$, the MSGs are, respec-

tively, $Cmc2_1$ (No. 36.172), $Cm'c'2_1$ (No. 36.176), and $Cm'c'_1$ (No. 36.174). The first forbids a net magnetization, whereas the latter two allow weak ferromagnetism while preserving coplanarity, in agreement with the DFT results. In these latter two cases, each MSG is therefore associated with two oriented spin (super)groups⁸¹ corresponding to the collinear and coplanar magnetic states: $C^1m^1c^12_1^\infty N^m 1$ and $C^1m^{2M}c^{2M}2_1^{mM \times N} 1$, respectively. Here, N defines the spin quantization axis in the collinear state, while M denotes the weak-ferromagnetic moment in the coplanar state, whose quantization axis is normal to the spin plane. These groups were identified using FINDSPINGROUP⁸².

The group-subgroup hierarchy provides the classification summarized in Table II: purple multipoles are allowed already by the collinear spin group and describe the nonrelativistic altermagnetic SML; orange multipoles are additionally allowed by the coplanar spin group and are associated with the weak-ferromagnetic state; and black multipoles require the full MSG and are purely relativistic. The complete transformation properties are given in Appendix A 1.

The only multipole allowed in the collinear spin-group limit is the quadrupole $Q_{yz}^{(2)}$, which accounts for the dominant spin polarization parallel to the Néel vector, $S \parallel N$, for all three Néel-vector orientations. Its characteristic momentum-space structure, illustrated in Fig. 4, exhibits nodal planes at $k_y = 0$ and $k_z = 0$, accounting for the d -wave character observed in the DFT spin textures.

For $N \parallel x$, weak ferromagnetism is not symmetry-allowed, and therefore there are no orange terms. The remaining black multipoles require the full MSG and therefore represent SOC-induced contributions. The Rashba-like in-plane spin texture is described by the dipoles $Q_x^{(1)}$ and $Q_y^{(1)}$, corresponding to S_y and S_x , respectively. In addition, the out-of-plane spin-polarization component S_z identified in the DFT calculations is described by the symmetry-allowed quadrupole $Q_{xy}^{(2)}$, with nodal planes at $k_x = 0$ and $k_y = 0$ as illustrated in Fig. 4.

For $N \parallel y$ and $N \parallel z$, weak ferromagnetism is symmetry allowed and is described by the orange monopole $Q^{(0)}$, corresponding to the spin-polarization components, S_z and S_y respectively. Interestingly, the reduced symmetry of the coplanar state also admits additional momentum-dependent contributions. In particular, the orange dipole $Q_y^{(1)}$, contributing to S_x , is already permitted by the coplanar spin groups. Therefore, although $Q_y^{(1)}$ produces a linear-in-momentum spin polarization, it is not a purely relativistic contribution: once the coplanar magnetic configuration is established, this dipolar component can exist even in the absence of SOC. Nevertheless, SOC is required to stabilize the weakly ferromagnetic configuration, and in this sense $Q_y^{(1)}$ can still be regarded as a relativistic contribution. The same symmetry reduction also admits the orange diagonal quadrupoles $Q_{xx}^{(2)}$, $Q_{yy}^{(2)}$, and $Q_{zz}^{(2)}$, corresponding to spin-polarization components

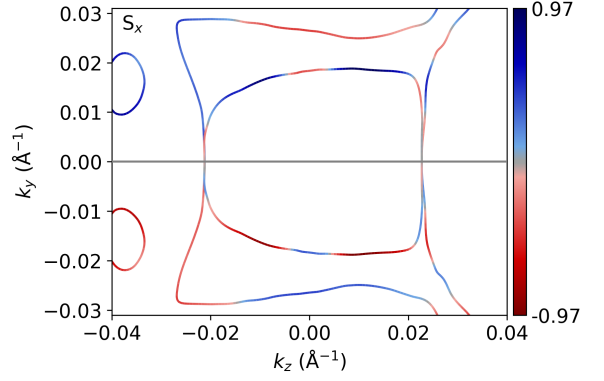


FIG. 5. Two dimensional constant-energy contours for $N \parallel x$ and spin-polarization component S_x , in the $k_x=0$ plane and at -1.7 eV from the top of the valence band, showing the p_y -wave spin-momentum locking character.

with quadratic momentum dependence proportional to k_x^2 , k_y^2 , and k_z^2 , respectively.

Finally, for these two Néel-vector orientations, the remaining black multipoles require the full MSG and therefore represent purely SOC-induced contributions. The quadrupoles $Q_{xz}^{(2)}$ and $Q_{xy}^{(2)}$ account for the additional in-plane spin-polarization component S_x found for $N \parallel y$ and $N \parallel z$, respectively.

Not all symmetry-allowed multipoles in Table II can be unambiguously identified in the DFT calculations of the previous section. Some may be too small to be resolved numerically, while others may be masked by larger contributions with similar momentum dependence. Nevertheless, the DFT spin textures are consistent with the multipolar structure predicted by the symmetry analysis.

Beyond identifying the individual multipolar contributions, the symmetry analysis also provides a direct interpretation of the topology of the relativistic spin texture. For the dominant spin-polarization component S_x for $N \parallel x$, the SML is given by the superposition $Q_{yz}^{(2)} + Q_y^{(1)}$. Consequently, the nodal plane $k_y = 0$, common to both multipoles, is preserved, whereas the nodal plane $k_z = 0$, belonging only to $Q_{yz}^{(2)}$, is no longer symmetry-protected. The resulting spin texture retains a single symmetry-protected nodal plane and therefore exhibits an effective p_y -wave character⁶, shown in Fig. 5, also referred to as a hybrid p/d -wave magnet^{83,84}.

By contrast, for the dominant spin-polarization component S_y for $N \parallel y$, the SML takes the form $Q_{yz}^{(2)} + Q_x^{(1)}$. In this case, neither of the nodal planes of the nonrelativistic contribution remains symmetry-protected, resulting in a spin texture without symmetry-enforced nodal planes^{84,85}.

Overall, these results demonstrate that combining symmetry analysis with first-principles calculations provides a systematic framework for disentangling the different microscopic contributions to the complex spin textures. As shown next, the same framework can be ex-

tended to spin-dependent response functions, establishing a connection with experiment.

E. Spin photogalvanic effect

The second-order dc spin photocurrent density driven by an optical electric field $\mathcal{E}(\omega)$ of frequency ω is

$$j^{s,a} = 2[\sigma_{\text{LP}}^{s,abc}(\omega) + \tau\eta_{\text{LP}}^{s,abc}(\omega)] \text{Re}[\mathcal{E}^b(\omega)\mathcal{E}^c(-\omega)] \\ + 2i[\sigma_{\text{CP}}^{s,abc}(\omega) + \tau\eta_{\text{CP}}^{s,abc}(\omega)] \text{Im}[\mathcal{E}^b(\omega)\mathcal{E}^c(-\omega)]. \quad (2)$$

Here, $s, a, b, c = x, y, z$, where s denotes the spin component of the current, a its propagation direction, and b, c the Cartesian components of the electric field. The real and imaginary terms in Eq. (2) describe the photogalvanic effect under linearly and circularly polarized (LP and CP) light, respectively. The photoconductivity, a rank-four tensor, is decomposed into two contributions, the shift and injection photoconductivities, $\sigma_{\text{LP}(\text{CP})}$ and $\eta_{\text{LP}(\text{CP})}$. The shift contribution is intrinsic, and therefore independent of scattering, whereas the injection contribution is proportional to the spin lifetime, denoted by τ . Physically, the shift current originates from inter-band coherence and corresponds to the spin-dependent displacement of the electronic wavepacket during an optical transition. By contrast, the injection current arises from an asymmetric photoexcitation of carriers in momentum space, resulting in a net spin current.

The symmetry-allowed photoconductivity elements are reported in Table III. They can be classified using the group-subgroup relations between the spin groups and MSGs established in Sec. IIID. As for the spin texture, this hierarchy separates components that survive in the collinear spin-group limit and are therefore of nonrelativistic altermagnetic origin, components allowed by the coplanar spin group and associated with weak ferromagnetism, and purely relativistic components that require the full MSG. Thus, the same color coding introduced for the spin multipoles is used in the table.

The four spin photoresponses can be further classified according to their behavior under time reversal. The spin LP shift and CP injection conductivities are odd under time reversal, whereas the spin LP injection and CP shift conductivities are even (see Appendix A 2 for their transformation properties in Jahn notation). The MSGs corresponding to the different Néel-vector orientations share the same spatial symmetry operations, but differ in whether these enter as unitary operations g or as antiunitary operations $g\mathcal{T}$, where $\mathcal{T}$ denotes time reversal. For the time-reversal-even responses, $\mathcal{T}$ introduces no additional sign in the transformation law, so that g and $g\mathcal{T}$ impose the same invariance condition. Consequently, changing the Néel-vector orientation does not change the set of symmetry-allowed tensor components. For the time-reversal-odd responses, instead, $\mathcal{T}$ introduces an additional sign, so that g and $g\mathcal{T}$ impose

different invariance conditions. Consequently, the set of symmetry-allowed tensor components can depend on the Néel-vector orientation.

The two time-reversal classes also differ in their spin-group decomposition: the time-reversal-odd responses retain contributions in the nonrelativistic collinear spin-group limit, whereas the time-reversal-even responses contain only contributions associated with the weak-ferromagnetic coplanar state and the full MSG. We examine the four photoresponses individually below.

1. LP shift conductivity

The symmetry-allowed elements of the spin shift conductivity for LP light are listed in Table III(a). The first-principles calculations reproduce all the symmetry-allowed elements for the three Néel-vector orientations. Figure 6(a)–(c) shows the calculated spectra for the easy-axis configuration $\mathbf{N} \parallel y$, while the results for $\mathbf{N} \parallel x$ and $\mathbf{N} \parallel z$ are reported in the SM.

We first consider the purple elements, whose spin polarization is parallel to the Néel vector and which represent the altermagnetic contribution. These elements remain finite in the collinear configuration when SOC is switched off in the DFT calculations (see SM), confirming their nonrelativistic origin. They also provide the largest spin shift response: for $\mathbf{N} \parallel y$, $\sigma_{\text{LP}}^{y,yyy}$ and $\sigma_{\text{LP}}^{y,yzz}$ reach approximately $8 \mu\text{A}/\text{V}^2$ and $6 \mu\text{A}/\text{V}^2$, respectively, near $\hbar\omega = 4.2 \text{ eV}$ [Fig. 6(b)]. These peaks occur in the ultraviolet, well above the fundamental absorption onset, and are consistent with optical transitions from the strongly spin-polarized valence bands at $-1.5 \lesssim E \lesssim -1.2 \text{ eV}$ to the conduction band near 2.8 eV . For both tensor elements, the spin current flows along y , parallel to both its spin polarization and the Néel vector.

We next consider the orange elements, whose spin polarization is parallel to the weak-ferromagnetic magnetization: along z for $\mathbf{N} \parallel y$ and along y for $\mathbf{N} \parallel z$. Consistent with their classification in the coplanar spin-group limit, these elements remain finite when SOC is switched off in the DFT calculations while the canted magnetic configuration is retained, but vanish when the canting and net magnetization are removed. Thus, although they do not explicitly require SOC once the canted state has been established, their physical realization relies on SOC to stabilize weak ferromagnetism. The largest of these elements, $\sigma_{\text{LP}}^{z,yzy}$, reaches approximately $3 \mu\text{A}/\text{V}^2$ near $\hbar\omega = 4.2 \text{ eV}$ [blue curve in Fig. 6(c)]. Remarkably, despite the small weak-ferromagnetic moment, this response remains comparable in magnitude to the dominant nonrelativistic contributions, showing that weak spin canting can generate sizable spin photocurrents.

Finally, the black elements require the full MSG and describe the purely relativistic contribution. For $\mathbf{N} \parallel y$, their spin polarization is along x , and they vanish when SOC is switched off, confirming their relativistic origin. They are also the smallest calculated elements: $\sigma_{\text{LP}}^{x,xzz}$

(a) LP shift conductivity, σ_{LP}

		Spin projection					
		S_x		S_y		S_z	
$\mathbf{N} \parallel$	x	xxy, yxx, yyy, yzz, zyz		xxx, xyy, xzz, yxy, zxz		xyz, yxz, zxy	
$\mathbf{N} \parallel$	y	xxx, xyy, xzz, yxy, zxz		xxy, yxx, yyy, yzz, zyz		xxz, yyz, zxx, zyy, zzz	
$\mathbf{N} \parallel$	z	xyz, yxz, zxy		xxz, yyz, zxx, zyy, zzz		xxy, yxx, yyy, yzz, zyz	

(b) CP injection conductivity, η_{CP}

		Spin projection					
		S_x		S_y		S_z	
$\mathbf{N} \parallel$	x	xxy, yzy		yxy, zxz		xyz, yxz, zxy	
$\mathbf{N} \parallel$	y	yxy, zxz		xxy, yzy		xxz, yyz	
$\mathbf{N} \parallel$	z	xyz, yxz, zxy		xxz, yyz		xxy, yzy	

(c) CP shift conductivity, σ_{CP}

		Spin projection					
		S_x		S_y		S_z	
$\mathbf{N} \parallel$	x	xxy,zyz		yxy, zxz		xyz, yxz, zxy	
$\mathbf{N} \parallel$	y	xxy, yzy		yxy, zxz		xyz, yxz, zxy	
$\mathbf{N} \parallel$	z	xxy, yzy		yxy, zxz		xyz, yxz, zxy	

(d) LP injection conductivity, η_{LP}

		Spin projection					
		S_x		S_y		S_z	
$\mathbf{N} \parallel$	x	xxy, yxx, yyy, yzz, zyz		xxx, xyy, xzz, yxy, zxz		xyz, yxz, zxy	
$\mathbf{N} \parallel$	y	xxy, yxx, yyy, yzz, zyz		xxx, xyy, xzz, yxy, zxz		xyz, yxz, zxy	
$\mathbf{N} \parallel$	z	xxy, yxx, yyy, yzz, zyz		xxx, xyy, xzz, yxy, zxz		xyz, yxz, zxy	

TABLE III. Symmetry-allowed elements of spin photoconductivities: (a) LP shift, (b) CP injection (c) CP shift, and (d) LP injection. Purple entries survive in the collinear spin-group limit, orange entries are additionally allowed in the coplanar spin-group limit, and black entries require the full MSG.

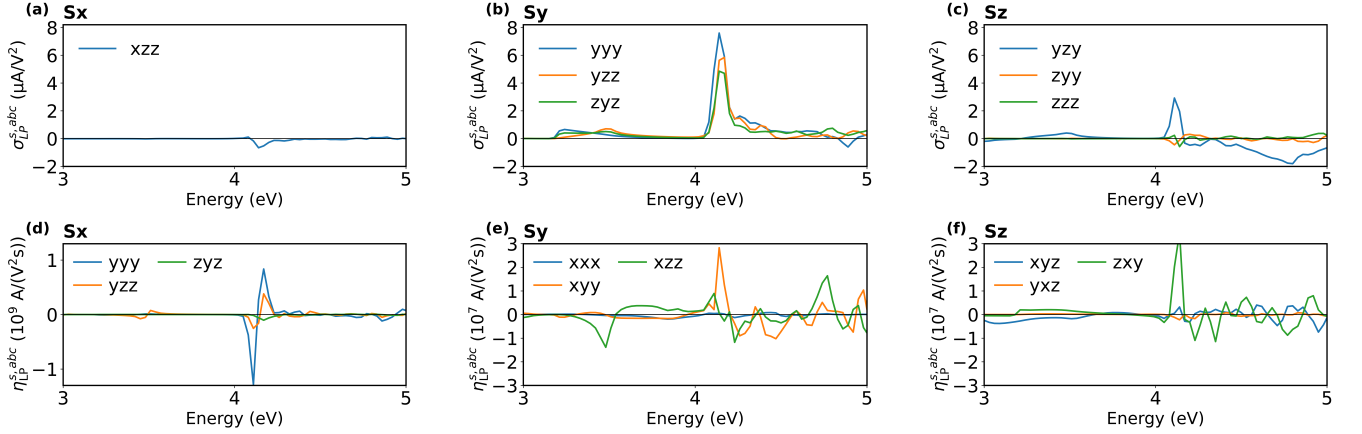


FIG. 6. Main components of the LP spin photoconductivity for $\mathbf{N} \parallel y$, resolved by spin-polarization direction. Panels (a)–(c) show the shift photoconductivity, while panels (d)–(f) show the injection photoconductivity. The three columns correspond to the spin polarizations S_x , S_y , and S_z , respectively. For the injection photoconductivity, note the different vertical-axis scale in panel (d) compared with panels (e) and (f).

reaches approximately $0.6 \mu A/V^2$ [Fig. 6(a)], about one order of magnitude below the largest nonrelativistic peaks.

The LP spin shift conductivity therefore reproduces the hierarchy identified in the equilibrium spin texture. The nonrelativistic altermagnetic contribution dominates, the contribution associated with weak ferromagnetism is smaller but remains sizable, and the purely SOC-induced contribution is substantially weaker.

2. LP injection conductivity

The symmetry-allowed elements of the LP spin injection conductivity are listed in Table III(d). Because this response is even under time reversal, the same elements are allowed for all three Néel-vector orientations, although their classification within the group-subgroup hierarchy depends on $\mathbf{N}$. Unlike the LP shift conductivity, the LP injection conductivity has no contribution in the collinear spin-group limit. It therefore separates into a coplanar sector (orange) and a purely SOC-induced

sector requiring the full MSG (black). For $\mathbf{N} \parallel x$, weak ferromagnetism is forbidden and the magnetic state remains collinear, so all allowed elements belong to the latter sector.

For $\mathbf{N} \parallel y$ and $\mathbf{N} \parallel z$, the two sectors can be distinguished by their spin polarization. Injection conductivity elements with spin component S_x are allowed in the coplanar spin-group limit and can be associated with the subdominant $Q_y^{(1)}$ dipole of the SML, whereas those with spin components S_y and S_z require the full MSG. The first-principles calculations confirm this classification: when SOC is switched off while retaining the coplanar magnetic configuration, the elements with spin component S_x remain finite, whereas the elements with spin component S_y and S_z vanish.

For the easy-axis configuration $\mathbf{N} \parallel y$, the calculated spectra are shown in Fig. 6(d)–(f). The response is dominated by the coplanar sector. In particular, $\eta_{LP}^{x,yyy}$, describing a spin current flowing along y with spin polarization along x under y -polarized light, reaches approximately $10^9 A/(V^2 s)$ near $\hbar\omega = 4.2$ eV [blue curve in Fig. 6(d)]. This value is approximately 30 times larger

than those of the largest purely SOC-induced elements, among which $\eta_{\text{LP}}^{y,xyy}$ and $\eta_{\text{LP}}^{z,zyz}$ reach peak values of approximately $3 \times 10^7 \text{ A}/(\text{V}^2\text{s})$ [Fig. 6(e),(f)].

Thus, whereas the LP shift conductivity is dominated by the nonrelativistic altermagnetic sector, the LP injection conductivity is dominated by the coplanar sector associated with the subdominant $Q_y^{(1)}$ dipole of the spin texture.

3. CP shift and injection conductivities

For CP light, the time-reversal parities of the shift and injection responses are interchanged relative to LP light. The time-reversal-even CP shift conductivity therefore follows the same group-subgroup hierarchy as the LP injection conductivity, whereas the time-reversal-odd CP injection conductivity follows that of the LP shift conductivity. Their symmetry-allowed elements are listed in Table III(c) and (b), respectively.

Accordingly, the CP shift conductivity has no contribution in the collinear spin-group limit. For $\mathbf{N} \parallel x$, all its allowed elements require the full MSG and are purely SOC-induced. For $\mathbf{N} \parallel y$ and $\mathbf{N} \parallel z$, elements with spin component S_x are instead allowed in the coplanar spin-group limit and are associated with the $Q_y^{(1)}$ dipole of the spin texture, while the remaining elements require the full MSG. The calculated response is dominated by this coplanar sector: $\sigma_{\text{CP}}^{x,zyz}$ reaches maximum absolute values of approximately $6 \mu\text{A}/\text{V}^2$ and $3 \mu\text{A}/\text{V}^2$ for $\mathbf{N} \parallel y$ and $\mathbf{N} \parallel z$, respectively. By comparison, the purely SOC-induced elements with spin components S_y and S_z reach only about $0.2 \mu\text{A}/\text{V}^2$.

The CP injection conductivity contains contributions from all three levels of the hierarchy. The collinear spin-group sector has spin polarization parallel to the Néel vector and represents the nonrelativistic altermagnetic contribution. For $\mathbf{N} \parallel y$ and $\mathbf{N} \parallel z$, the coplanar sector has spin polarization parallel to the weak-ferromagnetic moment, while the remaining elements are purely SOC-induced. The calculated response is dominated by the nonrelativistic sector: $\eta_{\text{CP}}^{s,xyy}$ and $\eta_{\text{CP}}^{s,zyz}$, with $s = x, y, z$ for $\mathbf{N} \parallel x, y, z$, respectively, reach peak absolute values of the order of $10^8 \text{ A}/(\text{V}^2\text{s})$. The coplanar and purely SOC-induced elements are approximately one order of magnitude smaller, with maximum absolute values of the order of $10^7 \text{ A}/(\text{V}^2\text{s})$. The complete CP spectra are reported in the SM.

Thus, the exchange of time-reversal parity between the shift and injection responses is reflected in their dominant contributions: the CP shift conductivity, like the LP injection conductivity, is dominated by the coplanar sector corresponding to the $Q_y^{(1)}$ dipole of the spin texture, whereas the CP injection conductivity, like the LP shift conductivity, is dominated by the nonrelativistic altermagnetic sector.

IV. DISCUSSION AND CONCLUSIONS

We have established a direct connection between the microscopic SML of the altermagnetic multiferroic BaCuF_4 and its nonlinear spin-photogalvanic response, as summarized in Fig. 7. By combining first-principles calculations with spin-group and MSG symmetry analysis, we have shown that the spin texture arises from the superposition of distinct spin multipoles with different microscopic origins.

The dominant nonrelativistic contribution is described by the quadrupole $Q_{yz}^{(2)}$, which produces a d -wave spin polarization parallel to the Néel vector. This contribution originates primarily from the rotation of the CuF_6 octahedra. Because this rotation is coupled to the polar distortion, the nonrelativistic SML reverses upon ferroelectric switching, realizing the altermagnetoelectric effect.

SOC introduces additional contributions whose form depends on the orientation of the Néel vector. For $\mathbf{N} \parallel x$, the magnetic state remains collinear, and the relativistic spin texture contains Rashba-like dipoles and additional quadrupolar components. For $\mathbf{N} \parallel y$ and $\mathbf{N} \parallel z$, SOC also induces spin canting and weak ferromagnetism, allowing further momentum-dependent multipoles. Importantly, some of these contributions, such as the dipole $Q_y^{(1)}$, are allowed already by the coplanar spin group. They do not explicitly require SOC once the canted configuration has been established, although SOC is required to stabilize that configuration. The momentum dependence of a spin texture alone is therefore insufficient to establish its microscopic origin, which instead follows from the hierarchy between spin-group and MSG symmetries.

This hierarchy is directly reflected in the spin-photogalvanic response. For the time-reversal-odd LP shift and CP injection conductivities, spin currents polarized along the Néel vector probe the nonrelativistic altermagnetic sector. In the weakly ferromagnetic configurations, currents polarized along the net magnetization identify the coplanar sector, while the remaining spin-polarization channels probe purely SOC-induced contributions. Their calculated magnitudes follow the same hierarchy: the nonrelativistic contributions are generally the largest, those associated with spin canting remain sizable despite the small weak-ferromagnetic moment, and the purely SOC-induced contributions are typically weaker. The time-reversal-even LP injection and CP shift conductivities provide complementary information and are dominated by the coplanar contribution associated with the $Q_y^{(1)}$ dipole of the spin texture.

Experimentally, the spin photogalvanic response could be detected by converting the optically generated spin current into a charge signal. In a BaCuF_4 /heavy-metal heterostructure, for example, a Pt, W, or Ta layer could convert the injected spin current into a transverse voltage through the inverse spin Hall effect. Rotating the Néel vector, reversing the ferroelectric polarization, or varying

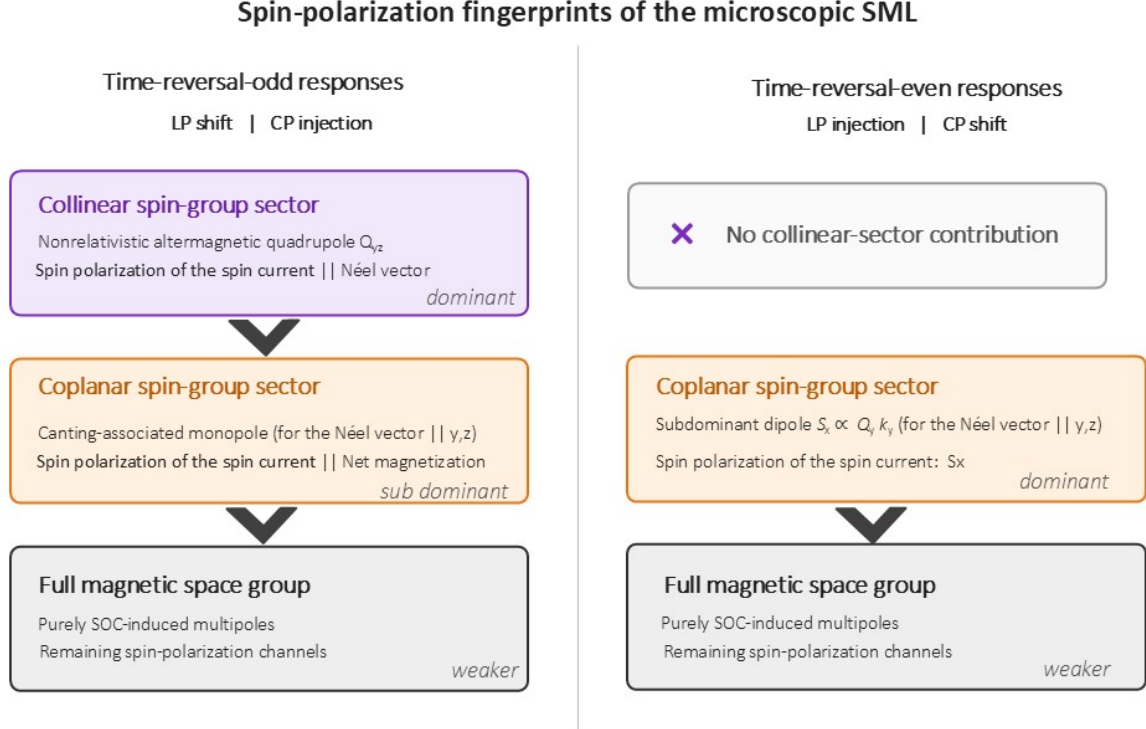


FIG. 7. Schematic relation between the microscopic contributions to SML and the spin-photogalvanic response. The time-reversal-odd LP shift and CP injection conductivities contain contributions from all three levels of the group-subgroup hierarchy. Spin currents with spin polarization parallel to the Néel vector probe the nonrelativistic altermagnetic quadrupole $Q_{yz}^{(2)}$; those polarized along the weak-ferromagnetic moment probe the canting-associated monopole $Q^{(0)}$; and the remaining spin-polarization channels probe purely SOC-induced multipoles. Their calculated magnitudes generally decrease in the same order. The time-reversal-even LP injection and CP shift conductivities contain no contribution from the collinear spin-group sector. For $\mathbf{N} \parallel y, z$, they are dominated by spin currents with spin component S_x , associated with the coplanar $Q_y^{(1)}$ dipole, whereas the remaining components require the full MSG and are purely SOC-induced.

the light polarization would allow spin currents polarized along the Néel vector, along the weak-ferromagnetic moment, and perpendicular to both to be distinguished. The measured voltage could thus provide an indirect but symmetry-resolved probe of the different microscopic contributions to the spin texture.

The group-subgroup framework developed here is not restricted to BaCuF_4 . It can be applied more generally to noncentrosymmetric altermagnets and multiferroics in which nonrelativistic altermagnetic SML coexists with Rashba-like, Weyl-like, or persistent relativistic spin textures and with deviations from collinear magnetic order. Our results therefore provide a general route for connecting the microscopic origin of complex spin textures to experimentally accessible nonlinear spin-transport responses.

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Appendix A: Tensor transformation properties in Jahn notation

1. Spin multipoles

The spin polarization transforms as an axial vector that is odd under time reversal, with Jahn symbol aeV , whereas the crystal momentum transforms as a time-reversal-odd polar vector. It follows that the multipolar coefficients $Q_{i\alpha\beta\gamma\dots}^{(n)}$ are characterized by the Jahn symbol $a^{n+1}eV[V^n]$. Accordingly, the zeroth-order (monopole) term transforms as the net magnetization, while the first-order (dipole) and second-order (quadrupole) terms are characterized by eV^2 and $aeV[V^2]$, respectively.

In this notation, each V represents a vector index, with the total number of V s specifying the tensor rank. Square brackets $[]$ denote symmetrization over the enclosed indices, whereas curly brackets $\{\}$ denote antisymmetrization. The symbols a and e encode additional transformation properties: a indicates odd parity under time reversal, while e denotes axial character.

These transformation properties determine the components of each spin multipole directly from the relevant spin group or magnetic space group, thereby providing a systematic framework for distinguishing nonrelativistic and relativistic contributions to the spin-momentum locking.

TABLE IV. Jahn symbols for the second-order spin photoconductivities.

conductivity	LP	CP
Shift	$aeVV[V^2]$	$eVV\{V^2\}$
Injection	$eVV[V^2]$	$aeVV\{V^2\}$

2. Second-order spin photoconductivity

The Jahn symbols of the second-order spin-photoconductivity tensors can be constructed directly from the transformation properties of their indices. The current direction is polar, whereas the spin polarization is axial, giving the factor eVV in the Jahn notation: the two V factors correspond to the current and spin-polarization indices, respectively, while e accounts for the axial character associated with the latter.

The two optical-field indices provide the remaining tensorial factor. For LP light, they enter symmetrically under interchange of their indices and therefore contribute $[V^2]$. For CP light, they enter antisymmetrically and contribute $\{V^2\}$. The tensorial structures of the LP and CP spin-photoconductivity tensors are thus $eVV[V^2]$ and $eVV\{V^2\}$, respectively.

The spin LP shift and CP injection conductivities are odd under time reversal and therefore carry the additional factor a , whereas the spin LP injection and CP shift conductivities are time-reversal even and do not. Together with the axial character and index symmetries discussed above, these time-reversal properties yield the Jahn symbols summarized in Table IV.

As for the spin multipoles, these transformation properties determine the symmetry-allowed spin-photoconductivity components. Comparing the constraints imposed by the nonrelativistic spin group and the corresponding magnetic space group then allows these components to be distinguished according to their non-relativistic or relativistic origin.

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Supplemental Material

S1. DENSITY OF STATES

Figure S1 shows the atom-projected density of states (DOS) of BaCuF_4 in the $Cmc2_1$ phase. The system is insulating, as also observed in the band structure discussed in the main text. The valence states are primarily derived from Cu d orbitals hybridized with F p -states, while Ba states lie well above the Fermi level and contribute mainly to the high-energy conduction region. The inequivalence of the spin-up and spin-down channels on the Cu sites reflects the local magnetic moments of the Cu^{2+} (d^9) ions. The elongation of the CuF_6 octahedra lifts the e_g degeneracy, as discussed in Sec. III-A of the main text. This results in a minority-spin gap of about 2.7 eV.

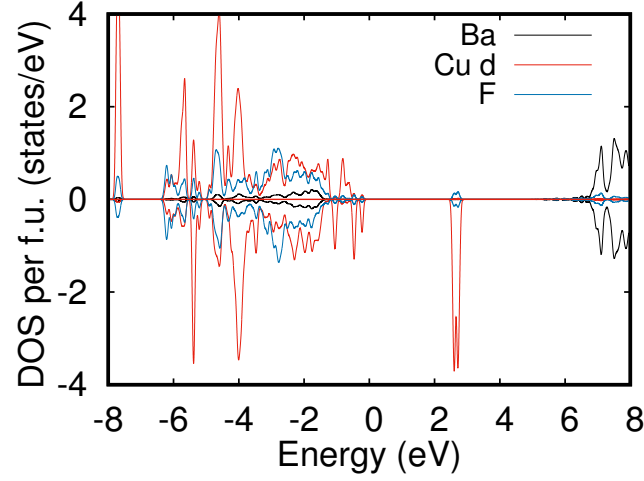


FIG. S1. Atom-projected DOS of BaCuF_4 in the $Cmc2_1$ ground state. Positive and negative values correspond to the spin-up and spin-down channels, respectively, and the energy is referred to the Fermi level. The DOS is projected onto Ba, Cu d -orbitals and F states.

S2. WANNIERIZATION PROCEDURE

We evaluated the second-order photocurrent responses following the methodology reported in Refs.^{27,36,50}. This approach is based on the interpolation of the DFT band structure via maximally localized Wannier functions (MLWFs)^{71,72}, as implemented in the WANNIER90 package⁷³. The Cu d and F p orbitals were taken into account, resulting in a total of 136 spinor bands in the presence of SOC. The wannierized bands are reported in Fig. S2 along the orthorhombic BZ path Γ -X-S-Y- Γ -Z-U-R-T-Z, both without and with the inclusion of SOC. The DFT bands are well reproduced in a wide energy range around the Fermi level, namely from -8 to 4 eV. As we can see from the two panels, SOC leaves the band gap essentially unchanged.

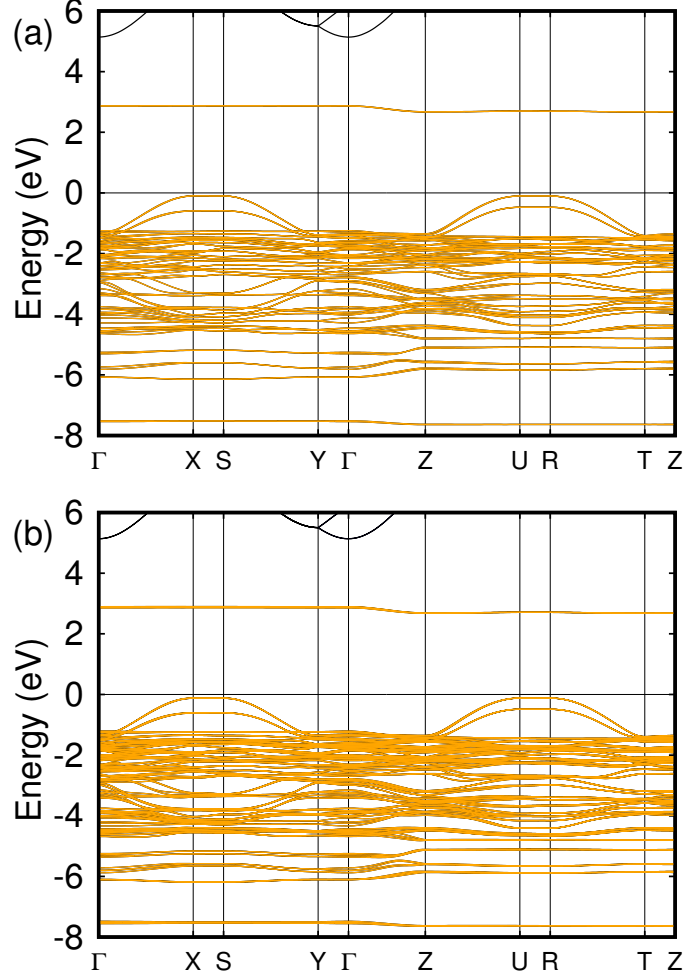


FIG. S2. DFT band structure (in black) and the Wannier-interpolated band structures (in orange) (a) without SOC and (b) with SOC. In all plots, the Fermi energy is set at 0 eV.

S3. SPIN-MOMENTUM LOCKING ALONG HIGH-SYMMETRY LINES OF THE BZ

Using DFT calculations and the multipole-extraction method described in Ref. [85], we calculate the magnetic multipoles in the reciprocal space. The results, summarized in Table S1, are consistent with the symmetry analysis presented in the main text.

The Rashba terms Q_y and Q_x , associated with the spin components S_x and S_y , respectively, are present for all three orientations of the Néel vector. The Q_{yz} term describes the nonrelativistic spin-momentum locking and appears along the diagonal of the table. Upon excluding the Rashba terms, the matrix is symmetric about its diagonal. The Q_0 term represents weak ferromagnetism, while the remaining terms describe quadrupolar contributions.

Below, we examine how the relativistic spin-momentum locking obtained from DFT simplifies along high-symmetry lines and for high-symmetry configurations.

		Spin components		
		S_x	S_y	S_z
N x		Q_{yz}, Q_y	Q_{xz}, Q_x	Q_{xy}
N y		Q_{xz}, Q_y	Q_{yz}, Q_x	Q_0
N z		Q_{xy}, Q_y	Q_0, Q_x	Q_{yz}

TABLE S1. Magnetization directions $N \parallel x, y, z$ for BaCuF₄ and corresponding spin-momentum locking for the spin components S_x, S_y and S_z . The dominant component has spin-momentum locking as the quadrupole Q_{yz} , while the subdominant components have the d-wave Q_{xz}, Q_{xy} and the s-wave Q_0 depending on the Néel vector orientation as reported in the Table. As in the main text, purple-coloured terms are those that would survive for collinear spin groups, while orange-coloured ones are those allowed by coplanar spin groups.

Table S2 summarizes the relativistic spin-momentum locking along the k_x axis.

		Spin components for $k_y=k_z=0$		
		S_x	S_y	S_z
N x		0	Q_x	0
N y		0	Q_x	Q_0
N z		0	Q_0, Q_x	0

TABLE S2. Magnetization directions $N \parallel x, y, z$ for BaCuF₄ and corresponding spin-momentum locking for the spin components S_x, S_y and S_z and with $k_y=k_z=0$.

For **N** || x , the S_y component exhibits Q_x symmetry [Fig. S3(a)], whereas for **N** || z , it contains both Q_0 and Q_x contributions [Fig. S3(b)].

For **N** || y , the S_y and S_z components exhibit Q_x and Q_0 symmetries, respectively [Fig. S4(a), (b)].

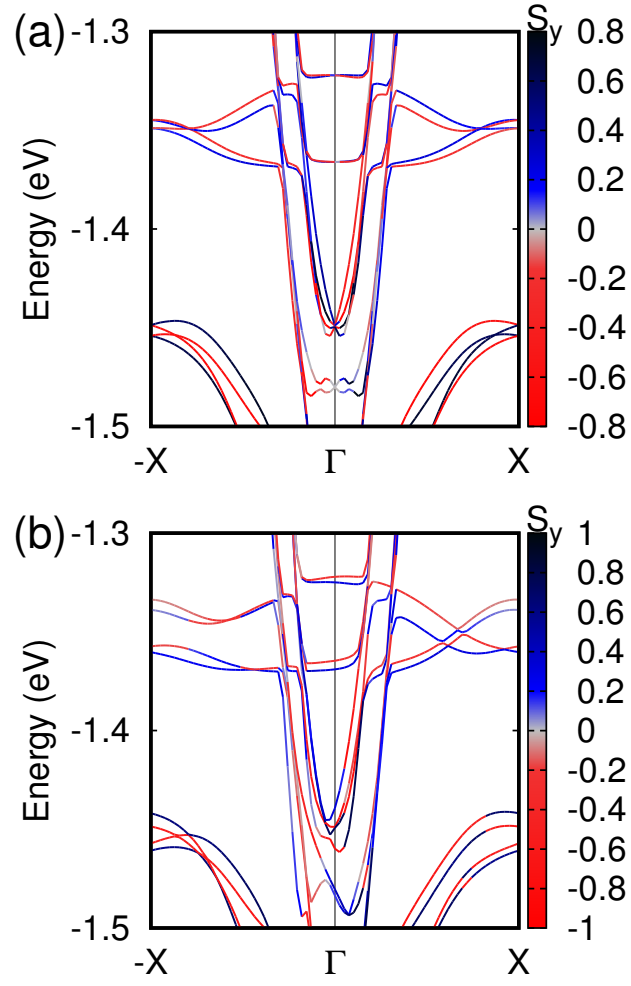


FIG. S3. Spin-resolved band structure for the Néel vector along the x- and z-axis, shown along the k_x -direction. The spin-momentum locking is Q_x and Q_0+Q_x for S_y , when the Néel vector is oriented along the x- and z-axis, respectively, as reported in Table S2.

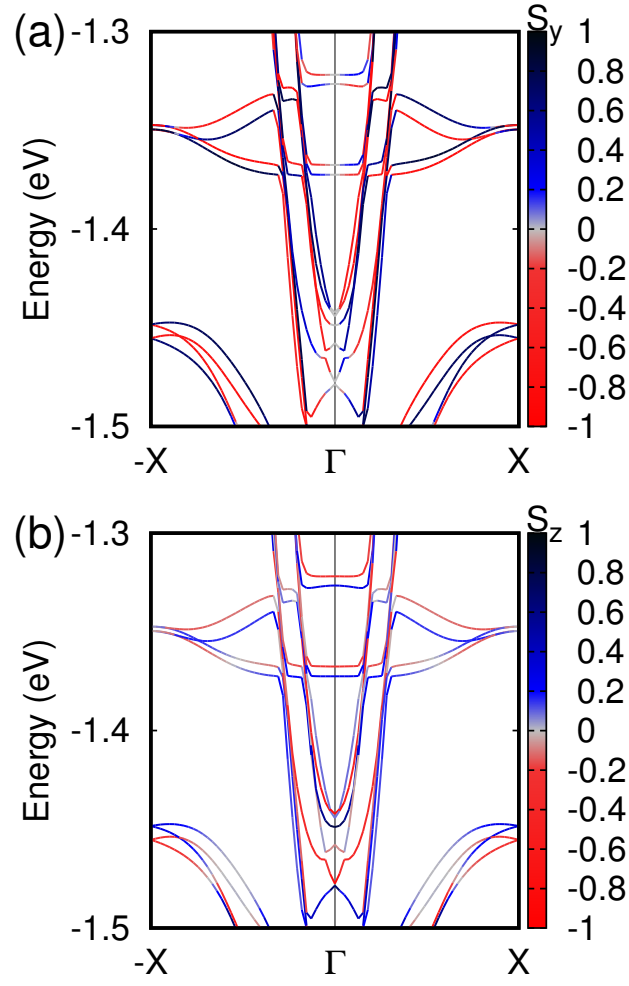


FIG. S4. Spin-resolved band structure for the Néel vector along the y-axis, shown along the k_x -direction. The spin-momentum locking is Q_x and Q_0 for S_y and S_z , respectively, as reported in Table S2.

Table S3 summarizes the relativistic spin-momentum locking along the k_y axis.

Spin components for $k_x=k_z=0$				
	S_x	S_y	S_z	
$\mathbf{N} \parallel x$	Q_y	0	0	
$\mathbf{N} \parallel y$	Q_y	0	Q_0	
$\mathbf{N} \parallel z$	Q_y	Q_0	0	

TABLE S3. Magnetization directions $\mathbf{N} \parallel x, y, z$ for BaCuF_4 and corresponding spin-momentum locking for the spin components S_x , S_y and S_z and with $k_x=k_z=0$.

For $\mathbf{N} \parallel x$ and $\mathbf{N} \parallel y$, the spin-momentum locking exhibits Q_y symmetry [Fig. S5(a), (b)].

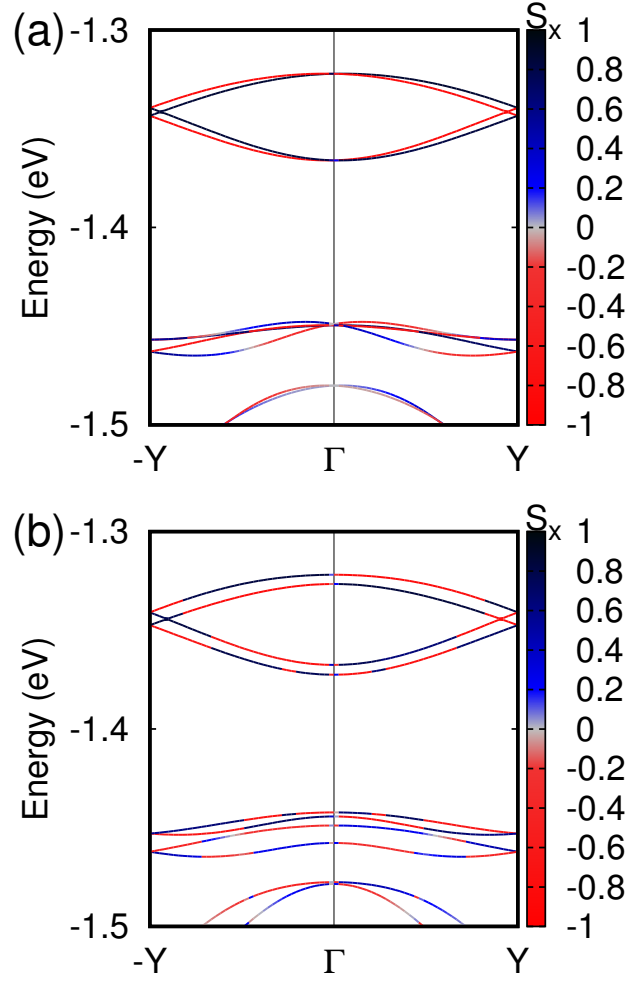


FIG. S5. Spin-resolved band structure for the Néel vector along the x- and y-axis, shown along the k_y -direction. For both case, the spin-momentum locking is Q_y as reported in Table S3.

For $\mathbf{N} \parallel z$, the S_x and S_y components exhibit Q_y and Q_0 symmetries, respectively [Fig. S6(a), (b)].

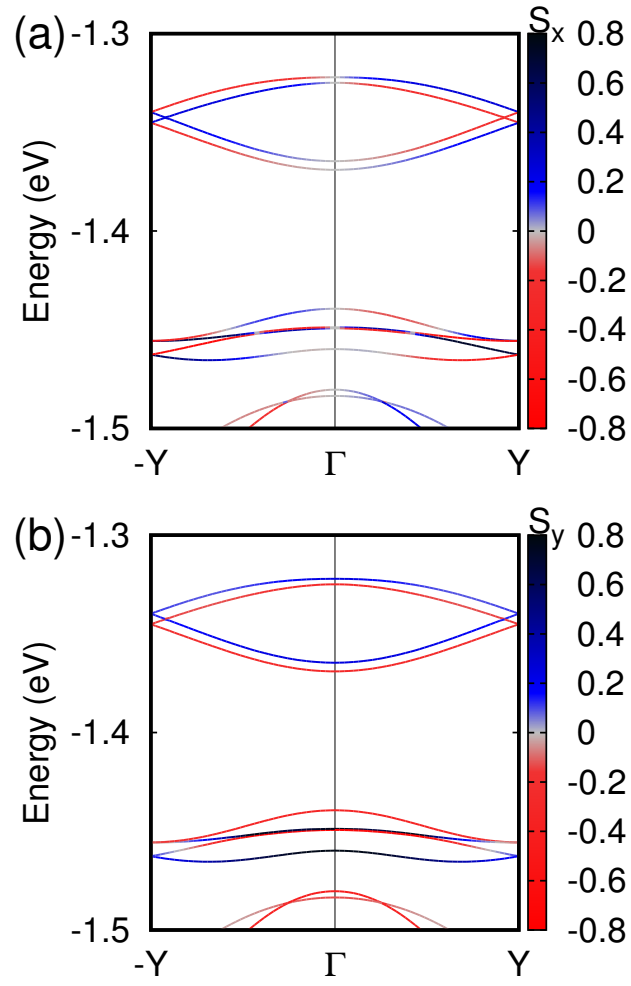


FIG. S6. Spin-resolved band structure along the z-axis, shown along the k_y -direction. The spin-momentum locking is Q_y and Q_0 for S_x and S_y , as reported in Table S3.

Table S4 summarizes the spin-momentum locking along the k_z axis, where only Q_0 components are present [Fig. S7].

Spin components for $k_x=k_y=0$			
	S_x	S_y	S_z
$\mathbf{N} \parallel x$	0	0	0
$\mathbf{N} \parallel y$	0	0	Q_0
$\mathbf{N} \parallel z$	0	Q_0	0

TABLE S4. Magnetization directions $\mathbf{N} \parallel x, y, z$ for BaCuF_4 and corresponding spin-momentum locking for the spin components S_x , S_y and S_z and with $k_x=0$ and $k_y=0$.

Along Γ -Z, the dominant spin component vanishes, and the band dispersion is slightly asymmetric about Γ , as seen in Fig. S7(a), (b).

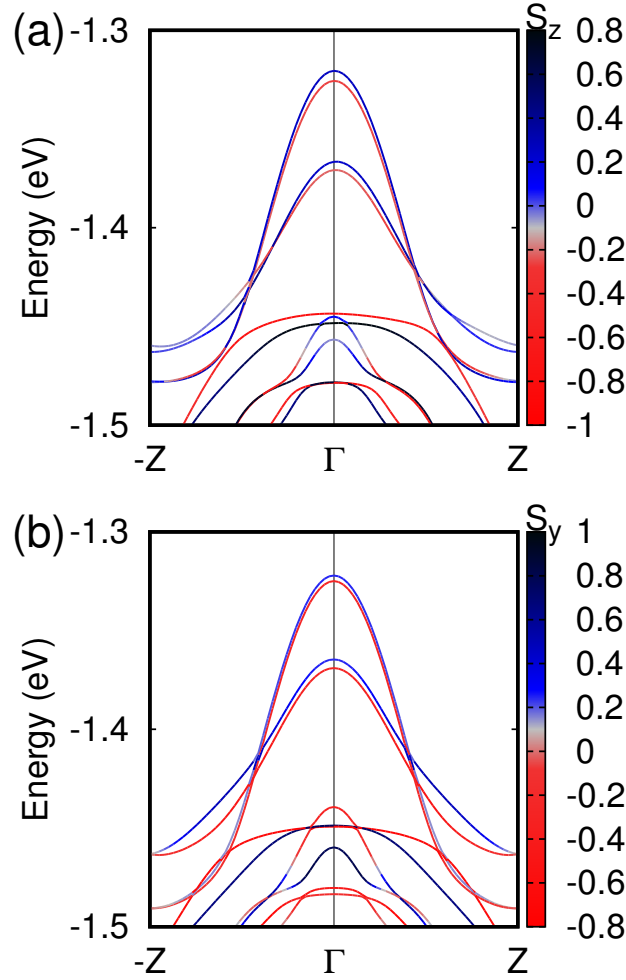


FIG. S7. Spin-resolved band structure for the Néel vector along the y- and z-axis, shown along the k_z -direction. For both case, the spin-momentum locking is Q_0 as reported in Table S4.

Finally, we report in Fig. S8 the case of the spin momentum locking along the diagonal (110) of the k-space; therefore, we have $k_x=k_y$ and $k_z=0$. Other directions were tested but not plotted. As a result of all tested directions, we obtained the quadrupole terms of the main text.

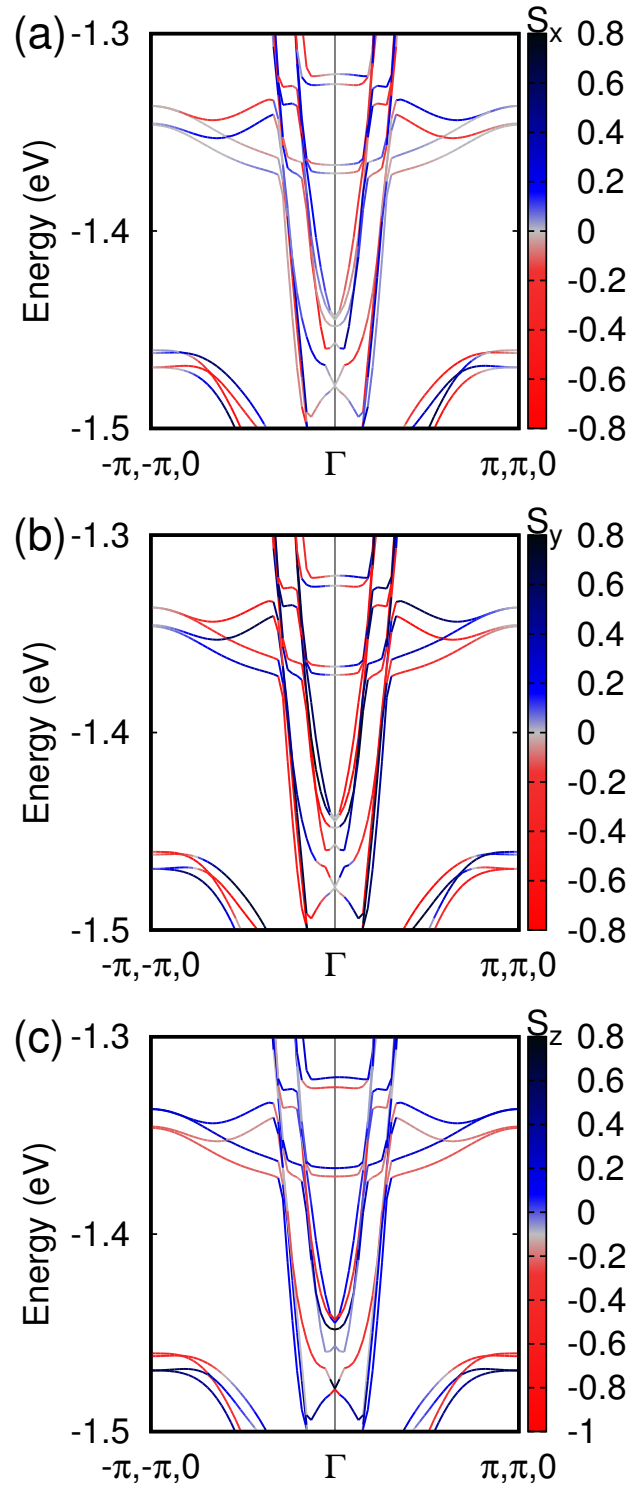


FIG. S8. Spin-resolved band structure for the Néel vector along the y-axis, shown along the diagonal for which $k_x=k_y$ and $k_z=0$. The spin-momentum locking is Q_x , Q_x and Q_0 for S_x , S_y and S_z , respectively.

S4. SPIN-RESOLVED FERMI SURFACE

For BaCuF_4 with $\mathbf{N} \parallel x$, the S_x , S_y , and S_z spin components exhibit Q_y , Q_x , and Q_{xy} symmetries, respectively, in the $k_z = 0$ plane (Table S1). This is confirmed by the constant-energy contours at 3.2 eV below the Fermi level, shown in Fig. S9. The colour scales indicate that S_x has the largest spectral weight, while S_y and S_z have comparable magnitudes. The S_y component exhibits Rashba-like spin-momentum locking, whereas S_z represents a subdominant relativistic contribution.

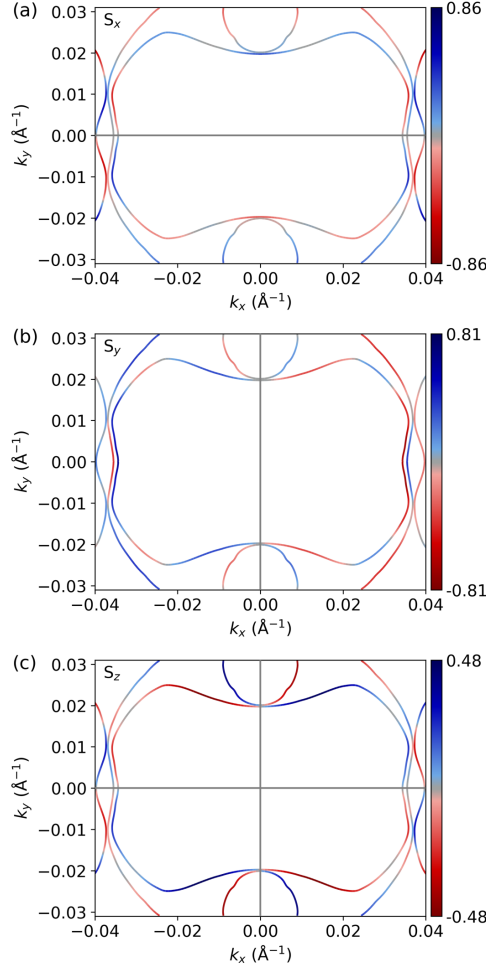


FIG. S9. Spin-resolved constant-energy contours near the centre of the Brillouin zone of BaCuF_4 for $\mathbf{N} \parallel x$, in the $k_z = 0$ plane at 3.2 eV below the Fermi level. Panels (a), (b), and (c) show the S_x , S_y , and S_z components, respectively. Solid black lines mark the intersections of the symmetry-protected nodal planes with the $k_z = 0$ plane. Red and blue indicate opposite spin polarizations, with the colour scales giving the spectral weight of each spin component.

S5. SPIN PHOTOGALVANIC EFFECT WITHOUT SOC

To calculate the spin photoconductivities without SOC, we follow Refs.^{27,36} and treat the two spin channels separately. Each spin photoconductivity is then obtained as the difference between the corresponding spin-up and spin-down conductivities, consistent with defining the spin photocurrent as $j^\uparrow - j^\downarrow$.

The LP spin shift and CP spin injection conductivities, shown in Fig. S10(a), (b), respectively, are the only nonzero contributions in the absence of SOC. The allowed tensor elements coincide with those permitted in the relativistic case for the spin parallel to the Néel vector. The dominant elements are σ_{LP}^{yyy} , σ_{LP}^{yzz} , and σ_{LP}^{zyz} for the LP shift conductivity, and η_{CP}^{xxy} and η_{CP}^{zyz} for the CP injection conductivity.

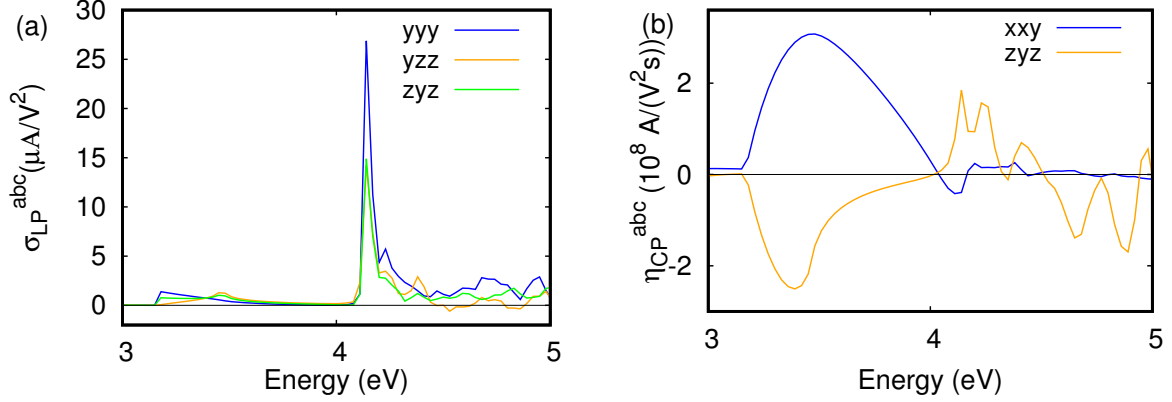


FIG. S10. Allowed components of the (a) LP spin shift photoconductivity and (b) CP spin injection photoconductivity without SOC.

S6. SPIN PHOTOCONDUCTIVITIES WITH SOC

Here, we show the main elements of all the spin conductivities for the Néel vector orientations $\mathbf{N} \parallel x$, $\mathbf{N} \parallel y$, and $\mathbf{N} \parallel z$.

A. LP shift conductivity

In Fig. S11 we report the main elements of the LP spin shift conductivity for the Néel vector orientations $\mathbf{N} \parallel x$ (a-c), $\mathbf{N} \parallel y$ (d-f), and $\mathbf{N} \parallel z$ (g-i) and for the spin components S_x , S_y and S_z . This response is odd under time reversal, and it shows different components depending on the Néel vector orientations. It shows nonrelativistic contributions and elements related to weak ferromagnetism and to relativistic effects. The hierarchy is the same for all the cases: the dominant contributions correspond to the spin along the Néel vector, the intermediate ones arise from spin canting, and the smallest are SOC-driven. By looking at Fig. S11 (a), (e), (i), we can see that the elements along the Néel vector direction are inherited from the nonrelativistic case, and the main contributions are $\sigma_{LP}^{s,yyy}$, $\sigma_{LP}^{s,yzz}$ and $\sigma_{LP}^{s,zyz}$, with $s = x, y, z$ for $\mathbf{N} \parallel x, y, z$, reaching approximately $8 \mu\text{A}/\text{V}^2$, $6 \mu\text{A}/\text{V}^2$ and $5 \mu\text{A}/\text{V}^2$ near $\hbar\omega = 4.2 \text{ eV}$, respectively. The elements presented in Figs. S11(f) and (h) originate from the weak ferromagnetism and are symmetry-allowed by coplanar spin groups, as shown in the symmetry analysis. The largest peaks for these contributions are reached by $\sigma_{LP}^{z,yzy}$ and $\sigma_{LP}^{y,yzy}$, for $\mathbf{N} \parallel y$ and $\mathbf{N} \parallel z$, respectively, of $3 \mu\text{A}/\text{V}^2$ near $\hbar\omega = 4.2 \text{ eV}$. They persist even in the absence of SOC when the weakly ferromagnetic configuration is enforced, indicating that SOC is only required to stabilize the noncollinear magnetic structure. When the weak-ferromagnetic component is switched off, these contributions do not survive. The remaining elements are activated by including SOC and are the smallest; they are the ones showed in Figs. S11 (b), (c), (d) and (g). Here we show only the largest contributions, which reach $0.2\text{-}0.5 \mu\text{A}/\text{V}^2$, around one order of magnitude smaller than the dominant nonrelativistic elements.

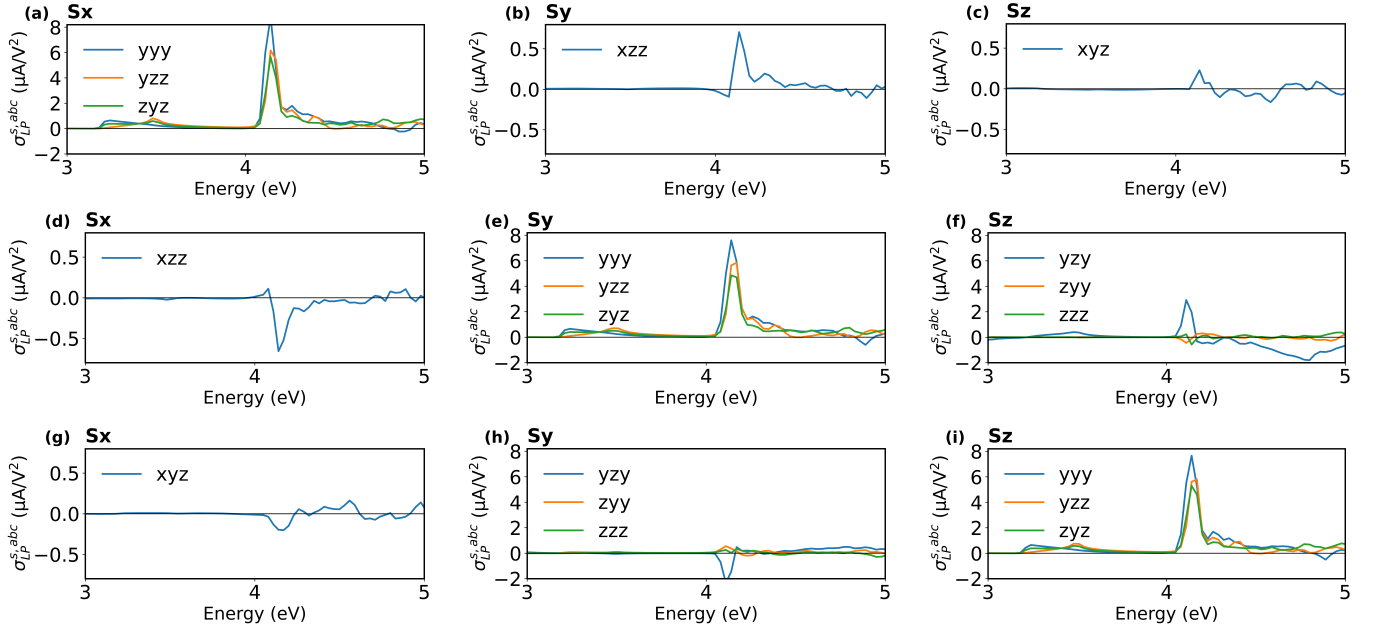


FIG. S11. Main components of the LP spin shift photoconductivity for (a-c) $\mathbf{N} \parallel x$, (d-f) $\mathbf{N} \parallel y$, and (g-i) $\mathbf{N} \parallel z$, resolved into spin components S_x , S_y , and S_z . Components of comparable magnitude are plotted using the same y -axis range, while different scales are used for components of different orders of magnitude.

B. LP injection conductivity

In Fig. S12 we report the main elements of the LP spin injection conductivity for the Néel vector orientations $\mathbf{N} \parallel x$ (a-c), $\mathbf{N} \parallel y$ (d-f), and $\mathbf{N} \parallel z$ (g-i) and for the spin components S_x , S_y and S_z . This response is even under time reversal, the same elements are allowed for all three Néel vector orientations, although their classification within the group-subgroup hierarchy depends on $\mathbf{N}$. Unlike the LP shift conductivity, the LP injection conductivity has no contribution in the collinear spin-group limit. It therefore separates into a coplanar sector and a purely SOC-induced sector. The elements related to the coplanar sector are the dominant ones, namely those for $\mathbf{N} \parallel y$ and $\mathbf{N} \parallel z$ with spin S_x . $\eta_{\text{LP}}^{x,yyy}$ reaches approximately $10^9 \text{ A}/(\text{V}^2\text{s})$ near $\hbar\omega = 4.2 \text{ eV}$ for $\mathbf{N} \parallel y$ (see Fig. S12 (d)), while $\eta_{\text{LP}}^{x,yzz}$ and $\eta_{\text{LP}}^{x,yyz}$ reach approximately $10^8 \text{ A}/(\text{V}^2\text{s})$ near $\hbar\omega = 4.2 \text{ eV}$ for $\mathbf{N} \parallel z$ (see Fig. S12 (g)). The remaining contributions, namely those purely relativistic, are much smaller than these ones, and are of the order of $10^7 \text{ A}/(\text{V}^2\text{s})$, as shown in the other panels of Fig. S12.

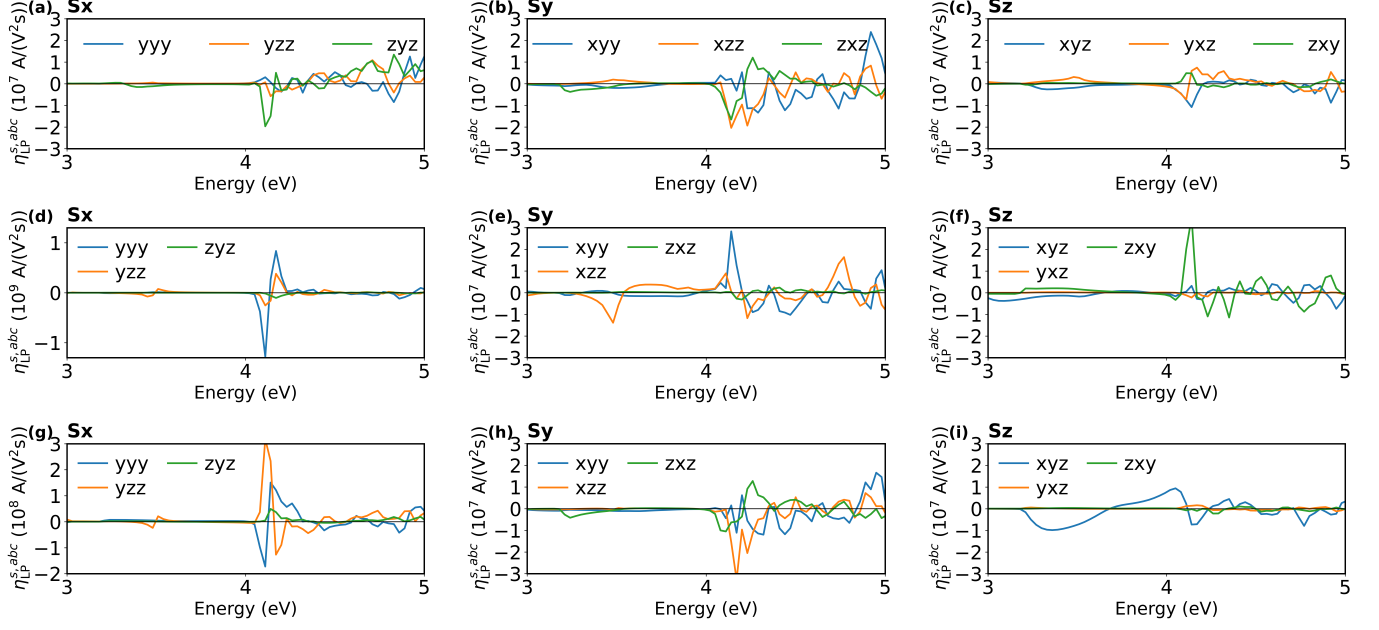


FIG. S12. Main components of the LP spin injection photoconductivity for (a–c) $\mathbf{N} \parallel x$, (d–f) $\mathbf{N} \parallel y$, and (g–i) $\mathbf{N} \parallel z$, resolved into spin components S_x , S_y , and S_z . Components of comparable magnitude are plotted using the same y -axis range, while different scales are used for components of different orders of magnitude.

C. CP shift conductivity

In Fig. S13 we report the main elements of the CP spin shift conductivity for the Néel vector orientations $\mathbf{N} \parallel x$ (a-c), $\mathbf{N} \parallel y$ (d-f), and $\mathbf{N} \parallel z$ (g-i) and for the spin components S_x , S_y and S_z . The CP shift conductivity is even under time reversal as the LP injection, therefore it shows the same behaviour, namely the same elements are allowed for all three Néel vector orientations, although their classification within the group-subgroup hierarchy depends on $\mathbf{N}$. It has no contribution in the collinear spin-group limit. It therefore separates into a coplanar sector and a purely SOC-induced sector. The elements related to the coplanar sector are the dominant ones, namely those for $\mathbf{N} \parallel y$ and $\mathbf{N} \parallel z$ with spin component S_x . $\sigma_{\text{CP}}^{x,xyz}$ reaches around $6 \mu\text{A}/\text{V}^2$ and $2 \mu\text{A}/\text{V}^2$ near $\hbar\omega = 4.2 \text{ eV}$ for $\mathbf{N} \parallel y$ and $\mathbf{N} \parallel z$, respectively, as shown in panels (d) and (g) of Fig. S13. The purely relativistic contributions are shown in the other panels and the largest peaks are one order of magnitude smaller.

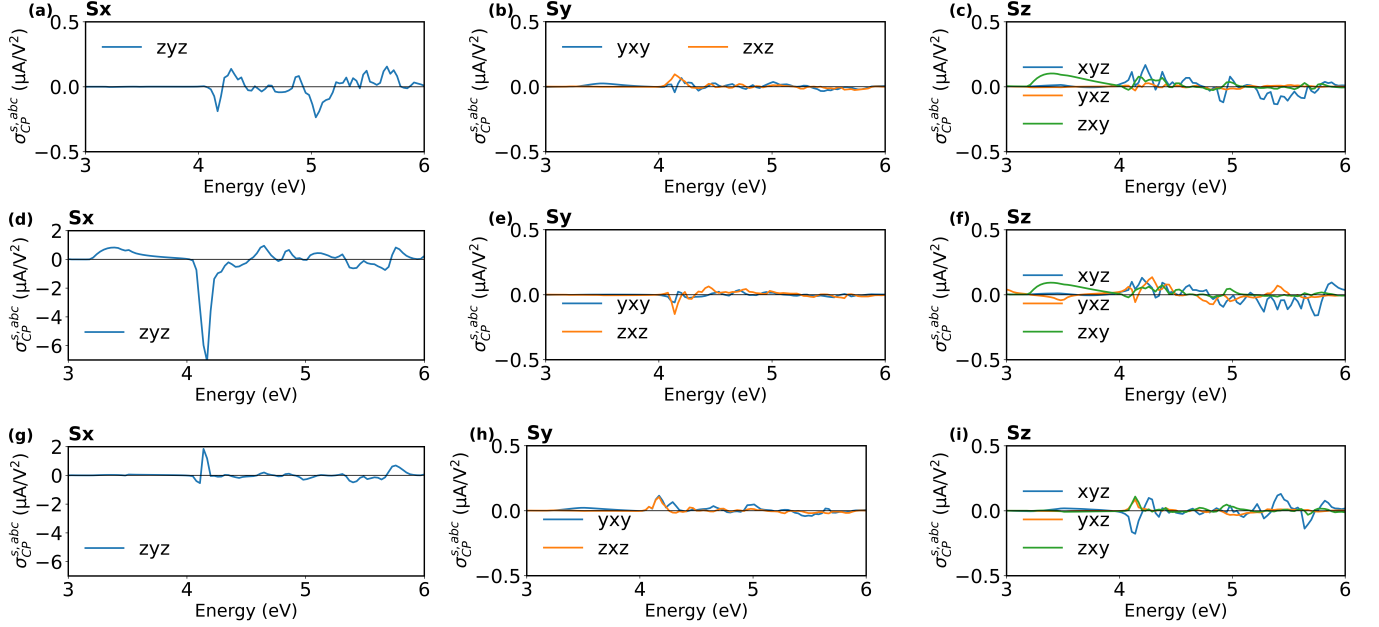


FIG. S13. Main components of the CP spin shift photoconductivity for (a-c) $\mathbf{N} \parallel x$, (d-f) $\mathbf{N} \parallel y$, and (g-i) $\mathbf{N} \parallel z$, resolved into spin components S_x , S_y , and S_z . Components of comparable magnitude are plotted using the same y -axis range, while different scales are used for components of different orders of magnitude.

D. CP injection conductivity

In Fig. S14 we report the main elements of the CP spin injection conductivity for the Néel vector orientations $\mathbf{N} \parallel x$ (a-c), $\mathbf{N} \parallel y$ (d-f), and $\mathbf{N} \parallel z$ (g-i) and for the spin components S_x , S_y and S_z . This response is odd under time-reversal and it behaves as the LP shift conductivity. It shows different elements depending on the Néel vector orientations and presents nonrelativistic contributions and elements related to weak ferromagnetism and to relativistic effects. By looking at Fig. S14 (a), (e), (i), we can see that the elements along the Néel vector direction are inherited from the nonrelativistic case, and the main elements are $\eta_{\text{CP}}^{s,xy}$ and $\eta_{\text{CP}}^{s,zy}$, with $s = x, y, z$ for $\mathbf{N} \parallel x, y, z$, respectively, reaching peak absolute values of the order of $10^8 \text{ A}/(\text{V}^2\text{s})$. The coplanar and purely SOC-induced elements are approximately one order of magnitude smaller, with maximum absolute values of the order of $10^7 \text{ A}/(\text{V}^2\text{s})$, as we can see from the other panels.

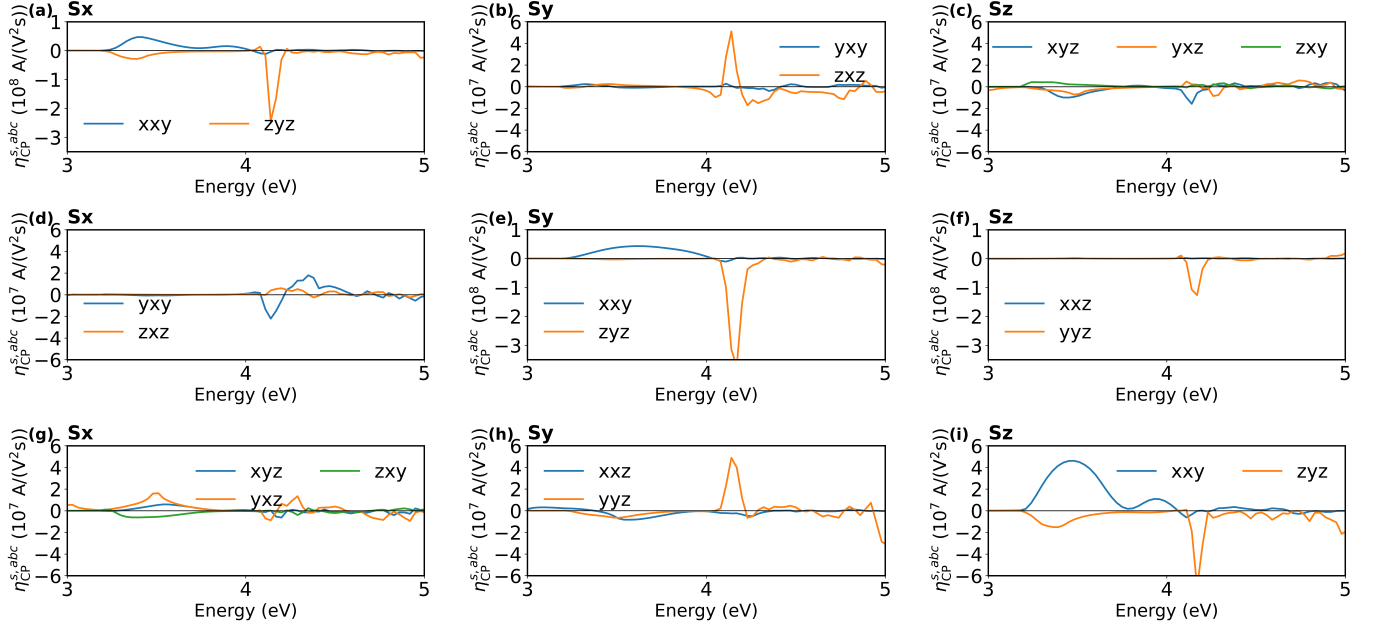


FIG. S14. Main components of the CP spin injection photoconductivity for (a–c) $\mathbf{N} \parallel x$, (d–f) $\mathbf{N} \parallel y$, and (g–i) $\mathbf{N} \parallel z$, resolved into spin components S_x , S_y , and S_z . Components of comparable magnitude are plotted using the same y -axis range, while different scales are used for components of different orders of magnitude.